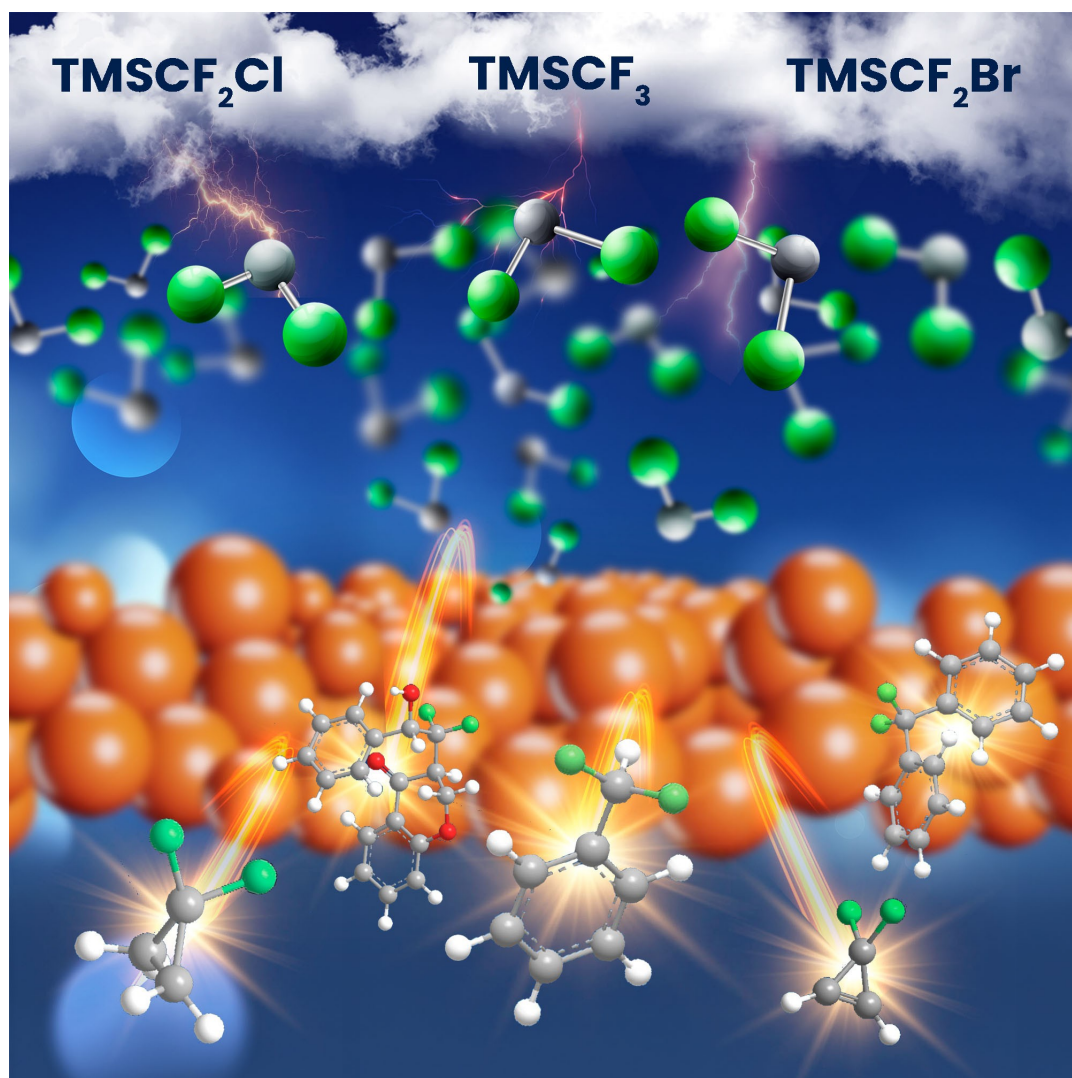


Advances in the Synthetic Utility of Difluorocarbene Generated from TMSCF_3 (Ruppert-Prakash Reagent) and Its Derivatives

Jagadeesh Krishnan,* Kaja Sahib*, Keerthana G. Nair*, Arshad Jouhar T, and Rony Rajan Paul*^[a]

This article is dedicated to the memory of Dr. Vijay Nair, whose inspiring contributions to the field of Organic Chemistry and unwavering dedication to research continue to influence and guide us. His legacy of excellence, curiosity, and generosity will always be remembered and cherished.



Abstract: Organofluorine compounds are of pivotal significance particularly, in drug and agrochemical industries and different strategies have been designed for their synthesis. The last two decades witnessed the emergence of difluorocarbene as an efficient synthetic tool, providing easy access to organofluorine compounds. This review summarises the reactions of difluorocarbene generated from Ruppert-Prakash reagent (TMSCF₃) and its derivatives TMSCF₂Cl and TMSCF₂Br. Among the various fluorination techniques available, the difluorocarbene chemistry offers a cost effective and easy procedure, opening avenue to a large number of organofluorine compounds. This review details the developments in the utility of difluorocarbene generated from TMSCF₃ and its derivatives, till date.

Keywords: Difluorocarbenes, Ruppert-Prakash reagent, TMSCF₃, TMSCF₂Br, TMSCF₂Cl, Utility of difluorocarbene

1. Introduction

Fluorine, being the 13th most abundant element in earth's crust shows its presence in nature in noticeable amounts. *Inter alia*, the scarcity of compounds with a C–F bond,^[1] can be attributed to the reduced nucleophilicity of fluoride under normal conditions and the low solubility of fluorine salts *viz.*, fluorite, fluorapatite, and cryolite.^[2] However, about 25–30 % of modern drugs (Figure 1)^[1b,2,3,4] as well as agrochemicals are endowed with fluorine atoms, as fluorinated molecules have improved membrane permeability, lipophilicity, and oxidation resistance. Also, an array of them is in clinical trials, owing to the fact that, fluorinated organic molecules have higher metabolic stability and bioavailability than their nonfluorinated equivalents.^[1b,2a] Also applications of fluorination in improving properties of materials are well documented.^[1a] In most of these molecules, fluorine exists as CF₃ group. Irrespective of the capability of the CHF₂ group to act as a hydrogen bond donor^[5] and the fact that the isosteric relationship amid the :CF₂ fragment and ethereal oxygens^[6] make difluoromethylated molecules suitable candidates in pharmaceutical research, compounds containing difluoromethylene fragment and difluoromethyl group are seen rarely.^[1–2]

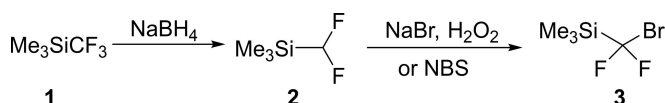
The available methodologies leading to gem-difluorinated compounds, include deoxofluorination,^[7] direct fluorination,^[8] transformation of the readily available gem difluorinated compounds,^[9] ring opening of difluorocyclopropanes synthesised from alkenes and alkynes^[10] and three component reactions involving difluorocarbene, electrophile, and a nucleophile.^[11] Among these strategies initiated by the difluorocarbene generated from Ruppert-Prakash reagent (TMSCF₃) and its derivatives such as TMSCF₂Br, and TMSCF₂Cl are of

great importance.^[2] In the last two decades, amongst the other fluorination strategies which require costly reagents and harsh conditions, trifluoromethyltrimethylsilane (TMSCF₃ or Me₃SiCF₃ or Ruppert-Prakash reagent) has emerged as a flexible reagent in organic synthesis as it is stable, cost-effective, and easy to handle.^[12–15]

Generally, difluorocarbene is a singlet carbene,^[16] basically electrophilic in nature, owing to the presence of a vacant p-orbital. Also, difluorocarbene (:CF₂) can be considered as one of the smallest carbon synthons containing fluorine atoms. It can install difluoromethyl (CHF₂–), fluoromethanylylidene (–CF=), difluoromethylene (–CF₂–), and difluoromethylidene (CF₂=) moieties into specific substrates.^[16a,17] An interesting general review on silicon based reagents has been published by Prakash *et al.* in 2017.^[2b] In this review we will focus on the synthetic applications of difluorocarbenes generated from TMSCF₃ **1** (Ruppert-Prakash Reagent) and its derivatives (TMSCF₂X) developed till date.

Among the difluorocarbene precursors, TMSCF₂X, Ruppert-Prakash reagent **1** is an easily accessible and stable reagent. It is also the least reactive among these precursors.^[18] (Bromodifluoromethyl)trimethylsilane (TMSCF₂Br) **3** is the most attractive reagent for the generation of difluorocarbene owing to its price and activity. TMSCF₂Br **3** can be prepared in 100 g scale from TMSCF₃ **1** reagent by reduction with NaBH₄ followed by bromination using NaBr and H₂O₂ or NBS (Scheme 1) and also it is commercially available.^[19,20]

TMSCF₂Cl **5** can be synthesised from TMSCF₂Br **3** *via* a high yielding halogen exchange reaction in presence of AgCl and n-Bu₄N⁺Cl[–] (Scheme 2).^[21]



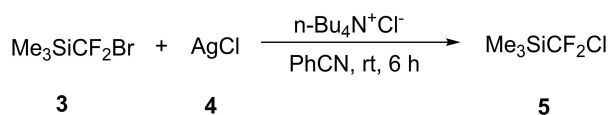
Scheme 1. Preparation of TMSCF₂Br.

[a] Dr. J. Krishnan, Mr. K. Sabib,* Ms. K. G. Nair,* Mr. A. J. T, Dr. R. R. Paul

Department of Chemistry, CMS College Kottayam (Autonomous), Kerala, 686001, India

E-mail: jagadeesh@cmscollege.ac.in
rony@cmscollege.ac.in

[*] Authors contributed equally.



Scheme 2. Preparation of TMSCF₂Cl.

2. Generation of Difluorocarbene

The difluorocarbene 7 generation from TMSCF₃ **1** may proceed by the treatment with NaI at temperatures above 80 °C,^[22] or under harsh conditions such as in the presence of LiI with triglyme as solvent at 140 °C or microwave condition at 170 °C.^[23] The simultaneous C–Si and C–F bond activation in the presence of TfOH to afford difluoromethylperfluorosulfonates proves the intermediacy of difluorocarbene from the mechanistic studies (Scheme 3).^[24]



Dr. Jagadeesh K. earned his Ph.D. in 2018 under the supervision of Prof. Vijay Nair at CSIR-NIIST Thiruvananthapuram. Following his doctorate, he was a postdoctoral research fellow at the Indian Institute of Science, Bangalore. He also served as a Research Scientist at Piramal Pharma Solutions, Hyderabad. Currently he is working as Assistant Professor in the Department of Chemistry at CMS College, Kottayam. His research interests include developing novel synthetic methodologies for biological applications, particularly in fluorine chemistry and tertiary amine catalysis.



Keerthana G Nair, born in 2001 in Kerala, India, completed her Bachelor's degree in Chemistry in 2022 and her Master's degree in Chemistry in 2024, both from Mahatma Gandhi University, Kottayam, Kerala. Following her post-graduation, she joined the research group of Dr. Rony Rajan Paul at CMS College, Kottayam, as a Research Assistant. Keerthana's academic journey reflects her passion for scientific exploration and dedication to the field of chemistry, making significant strides in research and laboratory expertise.



Arshad Jouhar T was born in 1999 in Kerala, India. He earned his Master's degree in 2022 from Mahatma Gandhi University Kottayam, Kerala, and his Bachelor's degree in 2020 from the University of Calicut, Kerala. He finished a six-month voluntary training program at the Phytochemistry Division of CMPR, AVS Kottakkal, Kerala. He is currently employed as a Project Associate under the supervision of Dr. Rony Rajan Paul.



Kaja Sahib was born in 2000 in Kerala, India. He received his B.Sc. degree in Chemistry from Mahatma Gandhi University, Kottayam, Kerala, in 2022, followed by a M.Sc. degree in Analytical Chemistry from the same university in 2024. Upon finishing his Master's degree, he joined Dr. Rony Rajan Paul's research group in the Department of Chemistry at CMS College (Autonomous) Kottayam, Kerala, India, as a Research Assistant, a position he continues to hold. His research interests are centred on exploring new reactions for organic synthesis.



Dr. Rony Rajan Paul received his PhD from CSIR-NIIST under the supervision of Professor Vijay Nair. He has been a visiting fellow at Universities of Potsdam and Duesseldorf, Germany and Nottingham Trent University, UK. After a post-doctoral stay with Professor Wim De Borggraeve at KU Leuven, Belgium he returned to his alma mater CMS College Kottayam as an Assistant Professor. Very recently he has been awarded SERB-SIRE fellowship at KU Leuven. His research interests mainly include development of new synthetic methodologies.

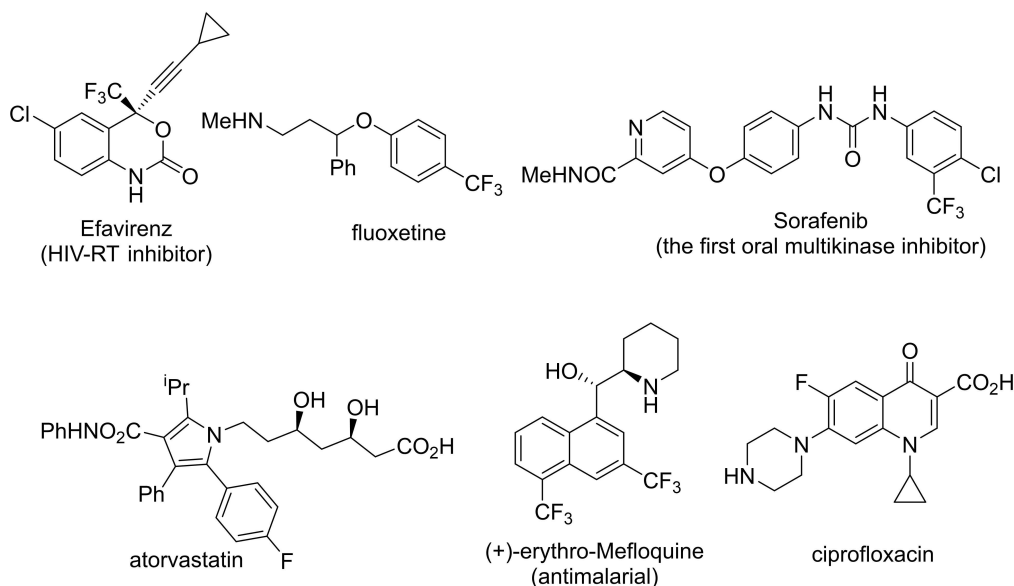
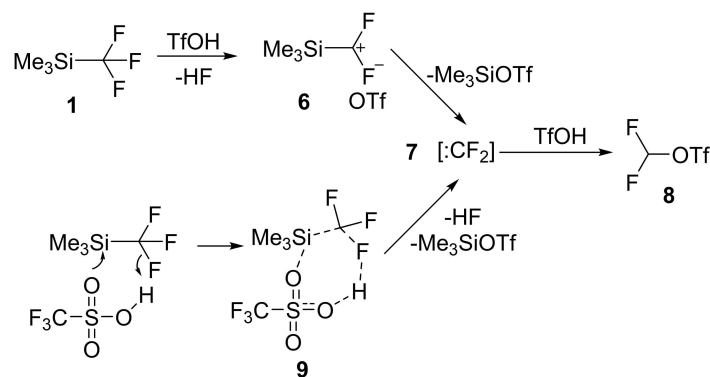
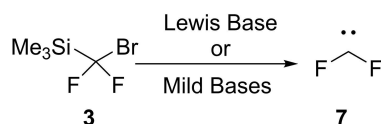


Figure 1. Fluorine containing drug molecules.



Scheme 3. Difluorocarbene generation from TMSCF₃.

Also, difluorocarbene can be generated from TMSCF₂Br **3** by the action of weak Lewis base.^[19c] The bromide and fluoride sources such as KF, TBAB, TBAF can also generate difluorocarbene **7** from TMSCF₂Br **3**. Hexamethylphosphoramide (HMPA) and N,N'-dimethylpropyleneurea (DMPU) are also known to initiate the formation of difluorocarbene **7** from TMSCF₂Br **3** at 0 °C (Scheme 4).^[25]



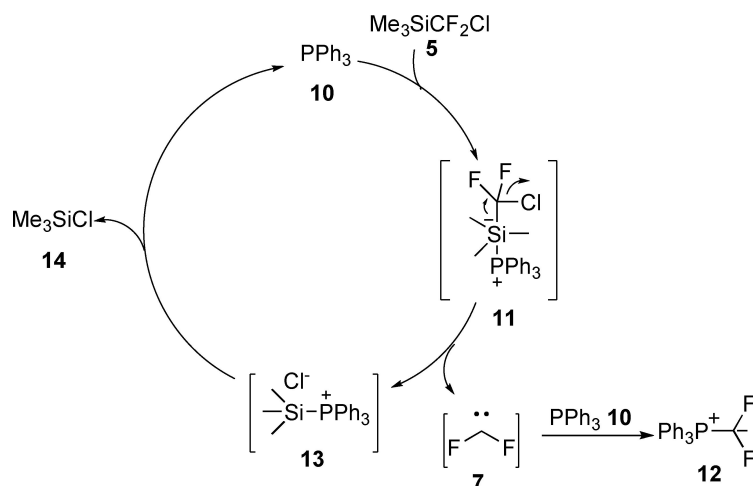
Scheme 4. Generation of difluorocarbene from TMSCF₂Br.

TMSCF₂Cl **5** can generate difluorocarbene **7** by the initiation of triphenylphosphine **10** via the activation of C–Si bond. Scheme 5 depicts the possible mechanistic route for the initiation.^[21]

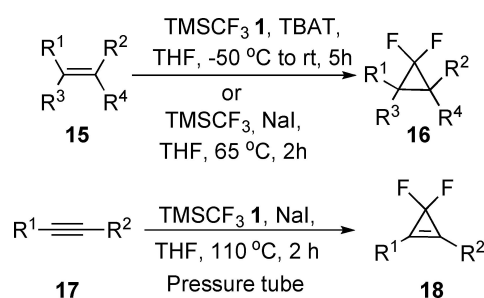
3. Reactions Involving Difluorocarbene Generated from TMSCF₃

3.1. Cyclopropanation and Related Reactions

Hu and Prakash were successful in the generation of difluorocarbene **7** from commercially available TMSCF₃ **1** at low temperature. Difluorocyclopropanes **18** and gem-difluorocyclopropanes **16** are achieved from alkynes **17** and alkenes **15** by this method (Scheme 6). TBAT is used to induce



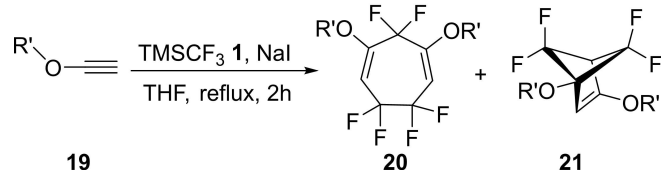
Scheme 5. Triphenyl phosphine initiated :CF_2 generation.



Scheme 6. Synthetic route to gem-difluorocyclopropanes/cyclopropenes.

decomposition of TMSCF_3 **1** to yield difluorocarbene **7** at low temperature. Meanwhile, it was showed that NaI plays the game changing role in the [2 + 1] cyclisation of alkynes **17** and alkenes **15** at higher temperature.^[22]

Inspired by the work of Hu and Prakash, O'Hagan *et al.* studied the reactivity of :CF_2 **7** generated from TMSCF_3 **1** with acetylene ethyl ether **19**. Interestingly, the reaction led to unexpected products **20** and **21** (Scheme 7).^[26] The seven membered rings **20** formed were found to be comparatively stable, whereas the latter bicyclic compounds **21** were found less stable.



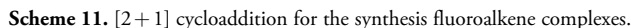
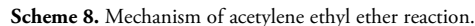
Scheme 7. Reaction of acetylene ethyl ether.

However, when conducted amidst a radical scavenger, TEMPO, the products **20** and **21** were not observed, proving that the reaction involves a radical mechanism as represented in the Scheme 8. The pathway A describes the formation of the seven membered product **20** and the pathway B illustrates the formation of bicyclic product **21**.

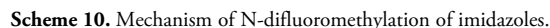
The mechanism of the formation of difluorocarbene **7** in the reaction is analogous to that reported by Prakash group in their N-difluoromethylation of benzimidazoles **25** and imidazoles **27** involving the Ruppert–Prakash reagent **1** under neutral reaction conditions (Scheme 9). These reactions were achieved under microwave irradiation conditions as well as conventional heating.^[23]

In this case, the difluorocarbene **7** is generated by reacting TMSCF_3 **1** and LiI. The reactions involving promoters CaI_2 , K_2CO_3 , KF, were found to be of low conversion. It was assumed that, the insoluble LiF present in the reaction enhanced the carbene generation and it resulted in improved yield of the N-difluoromethylated compound. The Li(I) cation traps the resulting fluoride ion, preventing its unwanted side reactions with the silyl group. The high boiling solvent triglyme allows to conduct the difluoromethylation reaction at high temperature safely. The reactions conducted with KF and NaF led to the complete utilization of TMSCF_3 **1** without yielding the product, presumably due to fast decomposition of TMSCF_3 **1** to CF_3H , carbene oligomers and some unwanted byproducts. The detailed study affirmed the vital role of Li(I) cation in tuning the fluoride availability along the course of the reaction (Scheme 10).

In 2014, Baker and co-workers reported the successful addition of difluorocarbene **7** and tetrafluoroethylene to the cobalt(I) metal to form cobalt difluorocarbenes **32** and $\eta^2\text{-TFE}$ complexes **33** (Scheme 11). In this reaction the difluorocar-



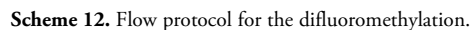
Scheme 9. N-Difluoromethylation of imidazoles and benzimidazoles.



benzene (:CF₂) **7** is generated using the Ruppert-Prakash reagent **1** as discussed earlier. The cobalt difluorocarbene **32** is formed

initially which subsequently undergoes a 2 + 1 cyclisation with difluorocarbene **7** to result in the TFE complex.^[27]

A flow protocol for the difluoromethylation of alkynes **17** and alkenes **15** in presence of the initiator NaI to generate difluorocarbene **7** was further developed (Scheme 12). The



catalytic amount of NaI and the controlled production of difluorocarbene **7** from TMSCF_3 **1** were made possible by the use of continuous flow technology. This flow reaction proceeded with 10 min residence time at a production rate of 1 mmol/min.^[28]

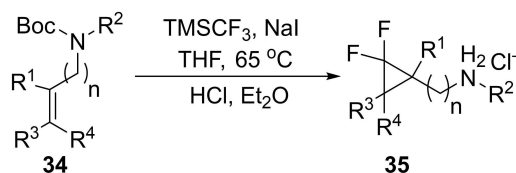
Grygorenko *et al.*, demonstrated a gram scale strategy for synthesizing gem-difluorocyclopropane decorated amines **35** from unsaturated N-Boc derivatives **34** by using TMSCF_3 **1** and NaI as initiator to generate difluorocarbene **7** (Scheme 13).^[29]

The order of reactivity of various amines are as shown in the Figure 2. It demonstrates that the process is directed by steric and electronic factors of the substrates **36** to **41**.

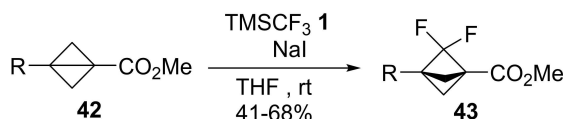
In 2019, Mykhailiuk and co-workers introduced a novel synthetic route to gem-difluorinated bicyclo [1.1.1] pentanes (F_2 -BCPs) **43**.^[30] The approach involved the addition of *in situ* generated $:\text{CF}_2$ -Carbene **7** to aryl/vinyl-substituted bicyclo [1.1.0] butanes **42** (Scheme 14). Investigation of various properties revealed the utility of obtained product **43** in mimicking benzene ring in pharmaceuticals. This novel approach therein allow entry to mono and bifunctional F_2 -BCPs – such as acid, amine, and alcohol.

3.2. Difluoroolefination of Diazo Compounds

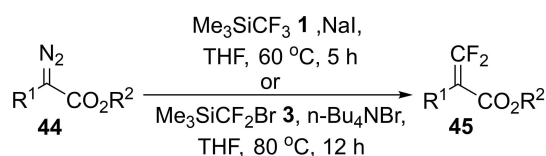
An interesting reaction to construct difluoroolefin **45** via the cross-coupling reaction of difluorocarbene **7** generated from TMSCF_3 **1** or TMSCF_2Br **3** with carbene fragment engen-



Scheme 13. Synthesis of amines bearing gem-difluorocyclopropane.



Scheme 14. Gem-difluorinated bicyclo [1.1.1] pentane synthesis.



Scheme 15. Synthesis of difluoroolefin from diazo Compounds.

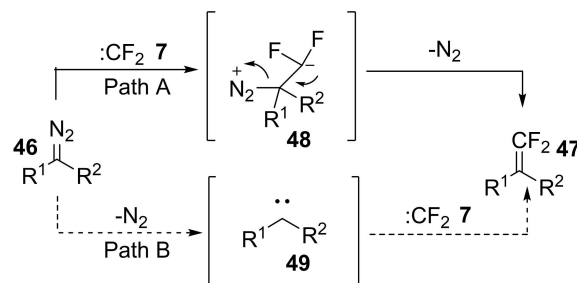
dered from the diazo compound **44** was reported by Hu's group.^[31] The generation of difluorocarbene from TMSCF_3 **1** involves NaI as the initiator, while the generation from TMSCF_2Br **3** involves $n\text{-Bu}_4\text{NBr}$ as the initiator both in THF at 60 °C and 80 °C respectively (Scheme 15).

The scope of diazo acetates **44** was tested under the optimum conditions, establishing the generality of reaction. The methodology was extended to various diaryldiazomethanes. Also, it was noticed that less stabilized diazo compounds like arylalkyldiazomethanes delivered cyclopropanation products by altering the ratio of diazirines and TMSCF_2Br **3** from 2:1 to 1:2.2. There are two plausible pathways for this reaction as shown in the Scheme 16. In path A, the mechanism is initiated by a nucleophilic addition of difluorocarbene **7** to the diazocompound **46** and subsequent elimination of N_2 . Path B explains the possibility of cross coupling of difluorocarbene **7** and the carbene generated from diazo compound *via* thermolysis.

3.3. Insertion Reaction

A novel synthesis of highly reactive $\text{C}_6\text{F}_5\text{CF}_2\text{CF}_2\text{Cu}$ and its treatment with several alkenyl, aryl and heteroaryl iodides was reported by Hu and co-workers in 2020.^[32] This work highlights the controlled introduction of two $:\text{CF}_2$ **7** groups into $\text{C}_6\text{F}_5\text{Cu}$ using TMSCF_3 **1**, without leading to a triple $:\text{CF}_2$ **7** introduction, even when excess TMSCF_3 **1** is used. The resulting $\text{C}_6\text{F}_5\text{CF}_2\text{CF}_2\text{Cu}$ species could further react with alkenyl, aryl and heteroaryl iodides, leading to various $-\text{CF}_2\text{CF}_2-$ inserted compounds **52**, thus providing an efficient and sustainable route to complex tetrafluoromethylene-bridged frameworks (Scheme 17). The possible mechanism suggests an initial formation of $\text{C}_6\text{F}_5\text{CF}_2\text{Cu}$ by the reaction of $\text{C}_6\text{F}_5\text{Cu}$ with highly reactive $\text{Cu}=\text{CF}_2$ generated from the Ruppert-Prakash reagent. Further, this intermediate reacts with $\text{Cu}=\text{CF}_2$ producing the double $:\text{CF}_2$ inserted $\text{C}_6\text{F}_5\text{Cu}$ molecule.

In 2021, Zhao and co-workers proposed a new synthetic strategy of gem-difluorinated 3-coumaranone **54**



Scheme 16. Mechanism of difluoroolefin synthesis from diazo Compounds.

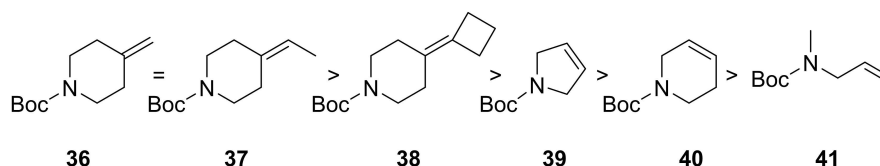
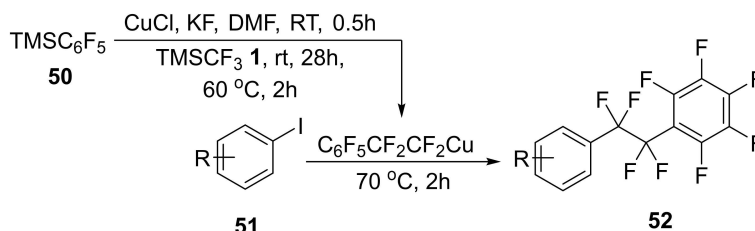


Figure 2. Relative reactivity of the substrates.

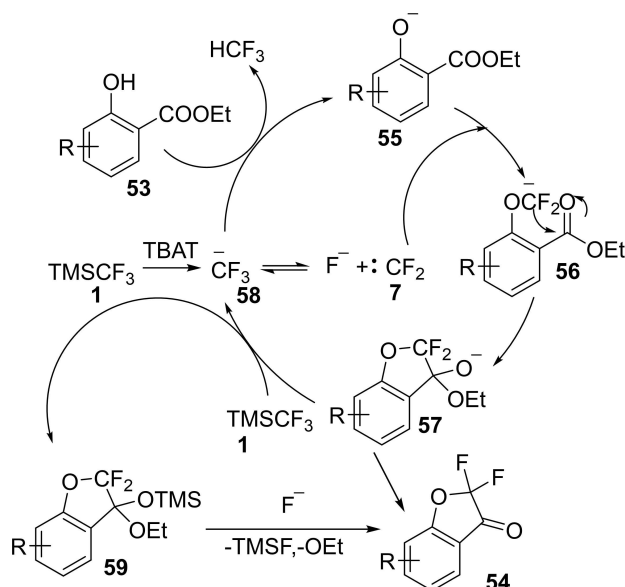


Scheme 17. Synthesis of tetrafluoroethylene-bridged structures.

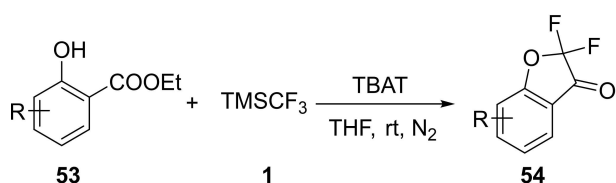
(Scheme 18).^[33] The synthesis was achieved through a :CF_2 7 mediated cyclization. The generation of :CF_2 7 from TMSCF_3 1 was facilitated by the activator, tributylammonium triphenyldifluorosilicate (TBAT); alternate fluorides like TBAF, KF, and CsF were used in lieu of TBAT, but failed to provide any desirable outcome. Mechanistic pathway of this approach explained the importance of :CF_2 7 in the cyclization reaction where it acts as difluorocarbene precursor, deprotonating agent as well as a silylating agent.

According to the mechanism, initially an *in-situ* generation of CF_3^- 58 took place, which deprotonated the substrate, leading to the formation of phenolate ion 55. The major step was the sequential addition of :CF_2 7 to phenolate ion and ester group of the substrate. Finally, products 59 and 54 were formed based on the consumption of TMSCF_3 1 (Scheme 19).

Later in 2023, Jones and Minshull investigated the reaction mechanism in detail.^[34] They found that location of electron-withdrawing substituent in the salicylate ring and the initial concentration of Ruppert-Prakash Reagent 1 have a significant impact in the net reaction rate.



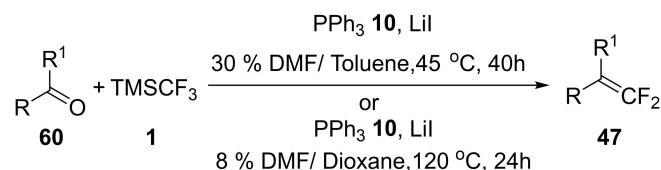
Scheme 19. Mechanism of the synthesis of gem-difluorinated 3-coumaranone.



Scheme 18. Synthesis of gem-difluorinated 3-coumaranone.

3.4. Deoxygenative Difluoromethylation

An adaptable one pot strategy of deoxygenative difluoromethylation of carbonyl compounds 60 was described by Prakash *et al.*, in 2016.^[35] This synthetic approach mainly involved three components, TMSCF_3 1, LiI, and PPh_3 10. The successful generation of difluorocarbene 7 was facilitated by Li^+ which also prevented TMSCF_3 1 from being decomposed. Moreover, the solvent combination enabled superior reactivity pattern, thereby decreasing reaction time and temperature (Scheme 20). The reliability of the reaction was examined by

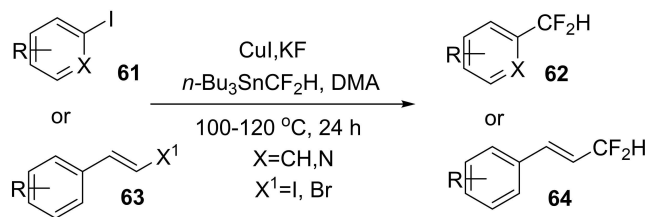


Scheme 20. Deoxygenative difluoromethylation of carbonyl compounds.

substituting LiI and PPh₃ with alternative reagents. Among the phosphines tested, electron-rich variants such as (Me₂N)₃P, intended to enhance the nucleophilicity of the ylide intermediate, and tributylphosphine, (n-Bu)₃P used to minimize steric hindrance, resulted in lower reaction yields. Additionally, replacing PPh₃ with dibenzothiophene or (PhO)₃P also did not provide desired outcomes. Moreover, control experiments revealed that the reaction did not proceed either in the absence of LiI/PPh₃ or when LiI was substituted with LiF or Ph₃PO.

3.5. Miscellaneous Reactions

In 2012, Prakash and co-workers came up with an innovative scheme for accessing tributyl(difluoromethyl) containing tin compounds from easily available difluorocarbene source-TMSCF₃ **1**.^[36] These compounds successfully facilitated difluoromethylation of iodoarenes **61** and β-styryl halides **63** in the presence of Cu catalyst. Furthermore, this catalytic difluoromethylation was extended to both activated and



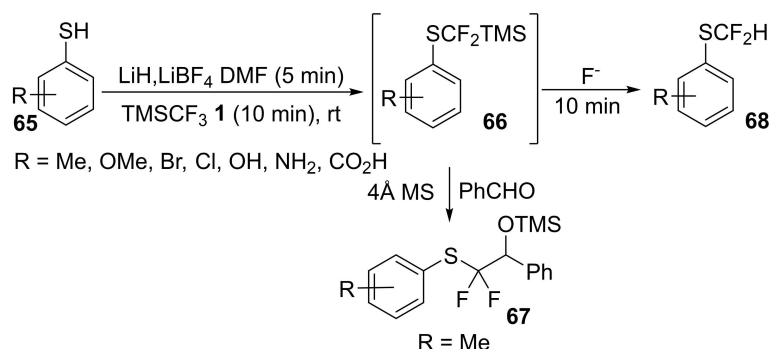
Scheme 21. Difluoromethylation of iodoarenes and β-styryl halides.

deactivated iodoarenes **61**, encompassing compounds with carbonyl substituents. This protocol provided greater yields even with β-styryl halides **63**, whose reactivity allowed bromides to be used to obtain the required products **62** or **64** in satisfactory yields. It is also noteworthy that the selection of solvent is crucial in this process, as the solvent molecules play the major role in stabilizing the Cu-CF₂H species (Scheme 21).

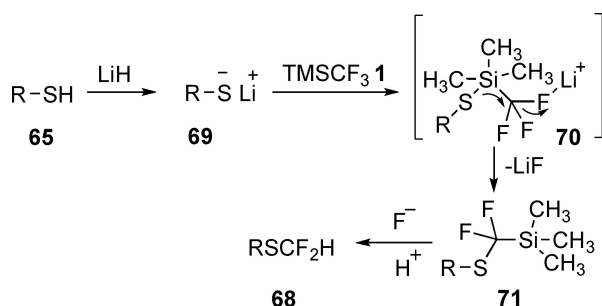
A facile synthetic approach of difluoromethyl sulfides **68** was introduced by Prakash and co-workers in 2015.^[37] Aliphatic and aryl thiols **65** underwent S-difluoromethylation in the presence of TMSCF₃ **1**, resulting in trimethylsilyldifluoromethyl sulfides **67**, which were further converted to difluoromethyl sulfides **68** by fluoride cleavage (Scheme 22). The fluoride abstraction was influenced by the cationic nature of lithium, with lithium salts containing weakly coordinated anions resulting in higher product conversions in the order OTf⁻ ≤ NTf₂⁻ < PF₆⁻ < BF₄⁻.

As per the proposed mechanism, lithium thiolate **69** formed by the reaction of thiol **65** and LiH, attack silicon and form a pentavalent intermediate **70**. The desired product **68** was finally formed by fluoride elimination of intermediate in the presence of Li⁺. Interestingly, when the synthesis was carried out using TIPSCF₃, no product generation took place due to the steric hindrance. These synthetic strategies have wide range of merits such as use of easily accessible starting materials, completion in shorter durations, simpler methods, and selectivity in presence of functional groups like -OH, -NH₂ and -COOH (Scheme 23).

In 2017, Hu and co-workers proposed a novel synthetic strategy of tetrafluoroethylene (TFE) by the dimerization of difluorocarbene **7** and investigated its utility in fluoroalkylation processes including pentafluoroethylation, tetrafluoroethylation and (aryloxy)tetrafluoroethylation.^[38] The reaction was carried out as a two-pot process, where TFE was generated from TMSCF₃ via the intermediacy of difluorocarbene by the action of NaI in THF as the solvent in one sealed tube, followed by the introduction of TFE into another sealed tube



Scheme 22. Synthesis of difluoromethyl sulfides.

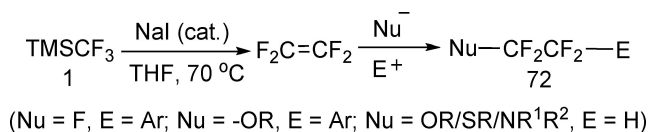


Scheme 23. Mechanism for the synthesis of difluoromethyl sulfides.

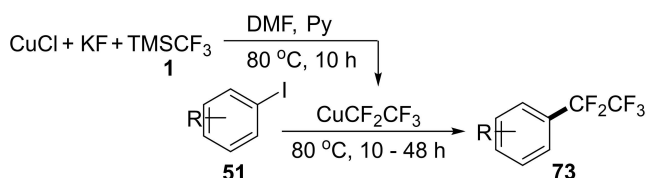
to promote fluoroalkylation. This approach is a facile entry to the generation of TFE in laboratory *via* safe and efficient protocol as well as it opens up a new methodology for the pentafluoroethylation (Scheme 24).

Also, they have introduced a new strategy of pentafluoroethylation using Ruppert–Prakash reagent **1** as a fluoroalkyl source.^[39] This synthetic approach was carried out through a C₁ to C₂ transformation wherein CuCF₃ generated from TMSCF₃ **1** was later converted into CuC₂F₅ which then took part in reactions with different aryl iodides **51** to form aromatic pentafluoroethyls **73**. This method thus explained the first successful strategy for pentafluoroethylation of aryl iodides **51** *via* homologation reaction by employing TMSCF₃ **1** (Scheme 25).

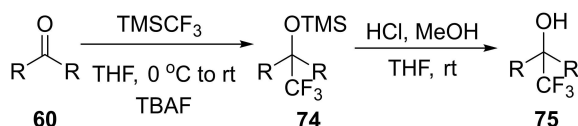
Jones *et al.*, conducted a detailed study of M⁺X[−] initiated transfer of CF₃ from TMSCF₃ **1** to aldehydes and ketones **60**



Scheme 24. Synthesis of tetrafluoroethylene.



Scheme 25. Synthesis of aromatic pentafluoroethyls.



Scheme 26. Trifluoromethylation of Ketones/Aldehydes.

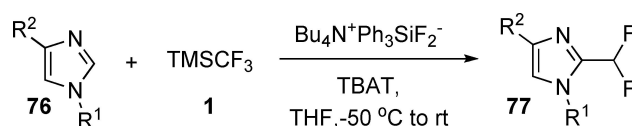
which included investigation of kinetics, selectivity and side reactions involved (Scheme 26).^[40]

Very recently, difluoromethylation of N-substituted imidazoles **76** was achieved by using the combination of TMSCF₃ **1** and TBAT by Grygorenko *et al.*, *via* the intermediacy of difluorocarbene.^[41] The reaction, carried out without any metals or strong bases, emerged as a convenient and reliable alternative to oxidative photochemical, and metal-catalyzed reactions of TMSCF₃ and similar reagents (Scheme 27). The NMR studies were in accordance with the predicted mechanistic pathway which involved the formation of corresponding 2-(trimethylsilyl)difluoromethyl-substituted derivative as intermediates. Theoretical study of reaction mechanism predicted two possible pathways of low activation barrier which involved the generation of imidazolidine anion as the intermediate.

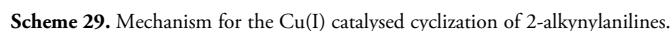
Recently, Qiu group demonstrated a Cu(I) catalyzed cyclization of 2-alkynylanilines **78** with TMSCF₃ and THF.^[42] This protocol provided an efficient and facile access to 2-haloalkoxy-3-alkyl(aryl)quinolines **79** and **80** in moderate to good yields with significant functional group compatibility and wide substrate scope (Scheme 28).

The proposed mechanistic pathway proceeds *via* the initial generation of difluorocarbene **7** from TMSCF₃ **1** in the presence of NaI and THF, which react with amino group of the motif. This is followed by the elimination of two hydrogen fluoride molecules resulting in the isonitrile **81**. Then the copper catalyst builds a π -dative bond with the alkyne, leading to intermediate **82**, which then eventually cyclizes to form copper complex **83**. In the next step, THF reacts with the copper complex **83**, forming intermediate **84**, which is then attacked by iodide ion at 80 °C, furnishing intermediate **85**. Protonation of intermediate **85** affords the desired product **79** and the catalyst gets into the catalytic loop. In the meantime, intermediate **86** is formed by the attack of difluorocarbene with intermediate **85** at 150 °C. Once again intermediate **86** is assaulted by iodide ion, followed by the addition of a proton, producing the observed product **80** and the catalyst is regenerated (Scheme 29).

In 2023, Davies and Sharland successfully developed a direct synthetic route for producing 2,2-difluorobicyclo[1.1.1]pentanes **89** and related bicyclic systems from α -allyldiazoacetate precursors **88**.^[43] This unique approach utilizes an intramolecular cyclopropane formation of



Scheme 27. Reaction of Azoles with the TMSCF₃-TBAT System.



88 $\xrightarrow[\text{THF, 65 } ^\circ\text{C}]{\text{Rh}_2(\text{Oct})_4, \text{DCM, rt, 45 min; TMSCF}_3 \text{ 1, NaI}}$ 89

R=H
 R=alkyl/phenyl

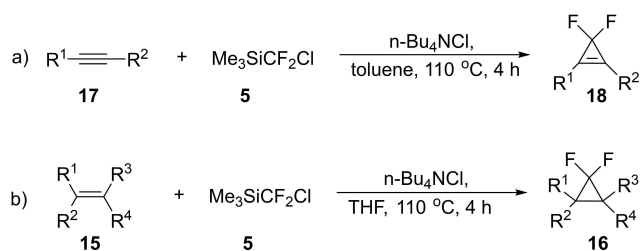
R=H
 R=alkyl/phenyl

Scheme 30. One-pot synthesis of 2,2-difluorobicyclo [1.1.1] pentanes.

4. Reactions Involving Difluorocarbene Generated from TMSCF₃Cl

The generation of difluorocarbene **7** from economical substrate, $\text{Me}_3\text{SiCF}_2\text{Cl}$ **5** under neutral and mild conditions to synthesise gem-difluorocyclopropenes **18** and gem-difluorocyclo-

clopropanes **16** from alkynes **17** and alkenes **15** via [2+1] cycloaddition reactions was another development in this line (Scheme 31).^[44] The difluorocarbene **7** are generated by the



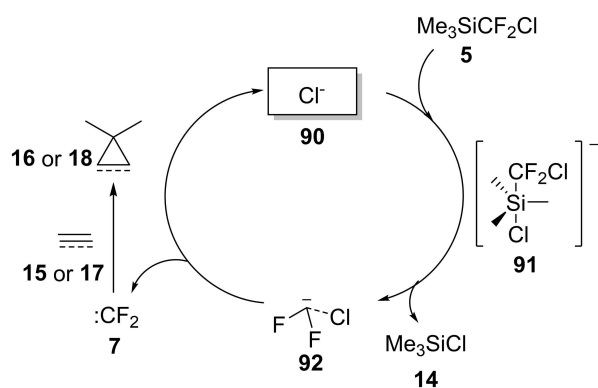
Scheme 31. Synthesis of gem-difluorocyclopropanes using $TMSCF_2Cl$.

addition of Cl^- to the Me_3SiCF_2Cl **5** and interestingly Cl^- is identified as the only source which allows the generation of difluorocarbene **7** at $110^\circ C$.

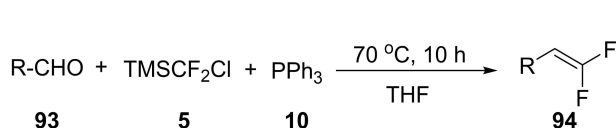
The reaction mechanism commences with the addition of Cl^- **90** to Me_3SiCF_2Cl **5** releasing Me_3SiCl **14** through a pentacoordinated silicon species **91** to form chlorodifluoromethylanion **92**. The anion then undertake an α -elimination of Cl^- **90** to deliver a singlet $:CF_2$ species **7** (Scheme 33). [2 + 1] cyclization reaction of the latter with alkynes **17** and alkenes **15** led to the formation of difluorocyclopropanes **18** and difluorocyclopropanes **16** respectively. Subsequently, the chloride ion **90** that is formed, enter into the next catalytic cycle, activating another Me_3SiCF_2Cl molecule **5** (Scheme 32).

4.2. Gem-Difluoroolefination

The gem-difluoroolefination of aldehyde **93** with $TMSCF_2Cl$ **5** in the presence of triphenylphosphine **10** was first of its kind



Scheme 32. Mechanism of gem-difluorocyclopropanes synthesis using $TMSCF_2Cl$.



Scheme 33. Synthesis of gem-difluoroolefination of carbonyl compounds.

when reported by Hu's research group.^[20] The reaction was proved to be general with various aromatic aldehydes **93** (Scheme 33). The reaction was also successful in aryl trifluoromethyl and chlorodifluoromethyl aromatic ketones.

This reaction is assumed to proceed through a difluoromethylenephosphoniumylide **13**, which is generated *in situ* by the reaction of difluorocarbene ($:CF_2$) **7** and triphenylphosphine **10** (Scheme 5). It is noteworthy that the novel synthetic strategy developed for the synthesis of $TMSCF_2Cl$ **5** from $TMSCF_2Br$ **3** via halogen exchange reaction in presence of $AgCl$ and $n-Bu_4N^+Cl^-$ is an environmentally benign method (Scheme 2).

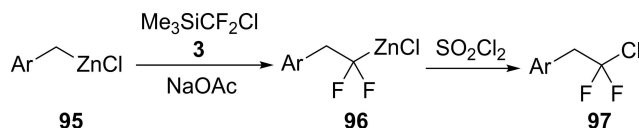
In 2014, Dilman and co-workers successfully synthesised compounds bearing the chlorodifluoromethyl group by employing organozinc reagents (Scheme 34).^[45] The reaction proceeds under the influence of sodium acetate where organozinc chloride **95** combine with (chlorodifluoromethyl) trimethylsilane **5** that acts as the $:CF_2$ source. Upon chlorination, the intermediate reagents lead to the chlorodifluoro-substituted products **97**. Interestingly, this methodology is only applicable to organozinc chlorides, as organozinc bromides yield unwanted bromodifluoro-substituted products through oxidation of bromide ions by the chlorinating agents. A similar transformation is achieved by the same group to obtain gem-difluorinated compounds from organozinc reagent by utilizing potassium bromodifluoroacetate serving as the $:CF_2$ source.^[46]

5. Reactions Involving Difluorocarbene Generated from $TMSCF_2Br$

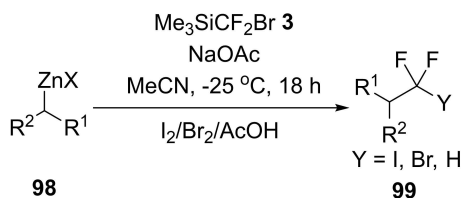
5.1. Insertion Reactions

In 2013, Dilman *et al.*, came up with a new strategy of introducing difluorocarbene **7** into a carbon-zinc linkage. Alkyl zinc /benzyl halides **98** were treated with difluorocarbene **7**, producing fluorinated organozinc compounds which on subsequent addition with various electrophiles (iodine, bromine, and proton) led to a variety of products **99** (Scheme 35).^[11]

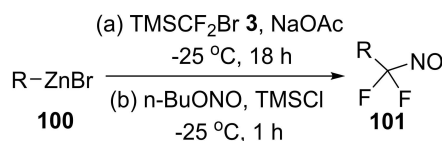
Subsequently, the same group synthesized gem-difluorinated nitroso compounds **101** by incorporating nitrosating electrophiles into fluorinated organozinc species (Scheme 36).^[47]



Scheme 34. Chloro difluoroiodomethylation of organozinc reagents.



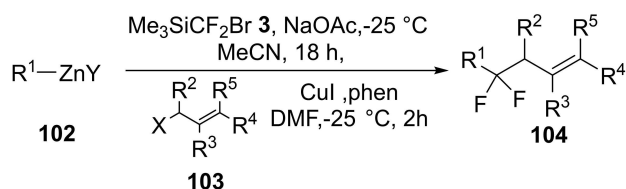
Scheme 35. Difluoroiodomethylation of organozinc reagents.



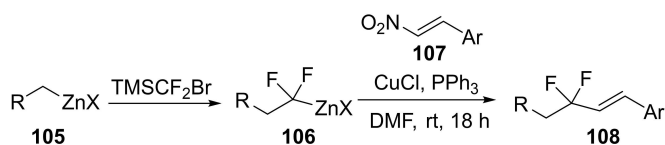
Scheme 36. Nitrosation of organozinc reagents.

Furthermore, with the aim of combining the efficiency of the described organozinc reagents **102**, Dilman *et al.*, developed a novel multicomponent reaction in 2014, synthesising different difluorocarbene containing compounds **104** via a three-component reaction of organozinc reagent **102**, difluorocarbene **7** and an allylic electrophile **103** (Scheme 37).^[48] The reaction proceeds by the initial :CF₂ **7** incorporation succeeded by a copper mediated allylation.

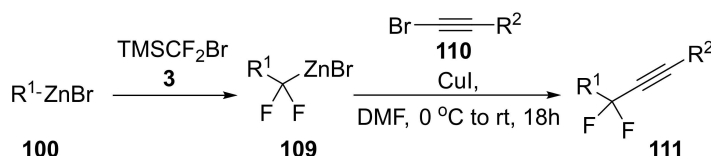
Shortly after developing an efficient method for generating gem-difluorinated organozinc reagents **106** via the intermedi-



Scheme 37. Allylation of organozinc reagents.



Scheme 38. Coupling of organozincs with β-nitrostyrene.



Scheme 39. Reaction of organozinc compounds with bromoalkynes.

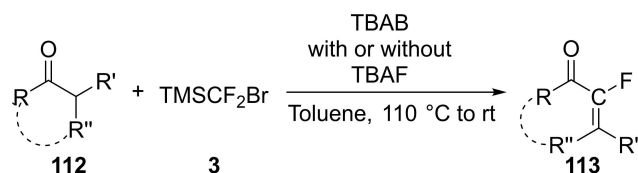
acy of difluorocarbene generated from TMSCF₂Br **3**, the same group reported a distinct copper(I) catalyzed reaction of difluorinated organozinc species **106** with β-nitrostyrenes **107**, wherein *E*-alkenes were formed as the sole products **108** (Scheme 38).^[49] The possible mechanism suggests a free radical pathway and it is also noticeable that the reaction advanced as conventional nitro group substitution *in lieu* of conjugate addition.

Also they put forward a simple entry towards difluorinated alkynes **111** via copper-catalyzed cross coupling reaction between 1-bromoalkynes **110** and gem-difluorinated organozinc compounds **109** generated using the previously proposed strategy (Scheme 39).^[50]

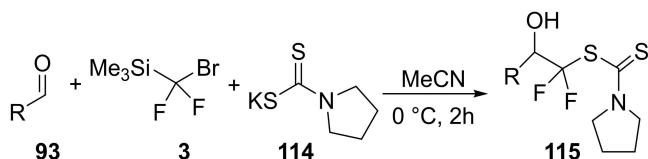
Wang group demonstrated the synthesis of α-fluoroenones and α-fluoroenals via a catalytic domino reaction. The reaction of aldehydes/ketones with TMSCF₂Br **3** resulted in the insertion of fluorine bearing unit to the α-C–C bond in the enolizable carbonyl compounds (Scheme 40).^[51] The reaction occurs *via* the coupling of difluorocarbene **7** (generated *in-situ*) and silylenol ether, forming α-carbonyl fluoroolefin **113**. *n*-Bu₄NBr in catalytic amount was employed in initiating the reaction and further catalytic domino procedures can be attributed to TMSBr formed as a side product, along with bromide and fluoride ions eliminated from TMSCF₂Br **3**. Aromatic ketones and aldehydes with electron-donating or electron-withdrawing substituents culminated in high yields of products.

Dilman and co-workers described a synthetic approach towards fluorinated silane containing a dithiocarbamate substituent **116** and validated its practical application in nucleophilic fluoroalkylation reactions of aldehydes and ketones.^[52] The forementioned silicon reagent **116** was obtained as a result of a simple substitution of bromine from (bromodifluoromethyl) trimethylsilane (Me₃SiCF₂Br) **3** by dithiocarbamate anion **114**. By engaging the above reagent with ketones/aldehydes **93**, corresponding alcohols **115** were afforded (Scheme 41).

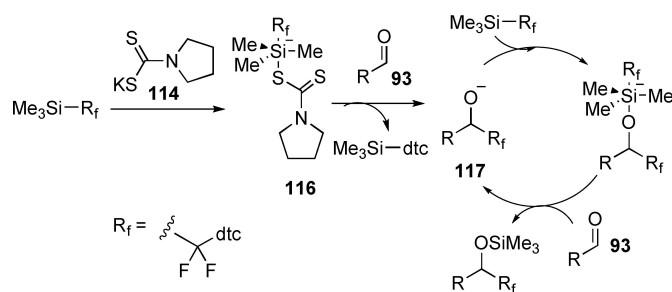
As per the proposed mechanism, the reaction involves the generation of alkoxide **117** which further drives the pathway (Scheme 42). The reaction is initiated by the activator potassium dithiocarbamate (K-dtc) **114** and it is noteworthy it does not progressed in the absence of an activator.



Scheme 40. Synthesis of α -fluoroenones and α -fluoroenals.



Scheme 41. Nucleophilic fluoroalkylation reactions of aldehydes.



Scheme 42. Mechanism of nucleophilic fluoroalkylation.

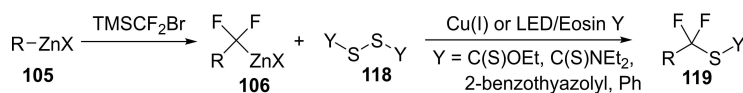
In 2016, Dilman's group reported coupling reactions of several disulfide reagents **118** with gem-difluorinated organozinc reagents **106**, which was synthesized using TMSCF_2Br **3** *via* the intermediacy of difluorocarbene^[53] These reactions were facilitated by using Copper (I) catalyst or by irradiating with visible light amid Eosin Y (Scheme 43). The reaction proceeds *via* generation of thiyl radical by single electron reduction of disulfide with light activated Eosin Y. Subsequently radical substitution occurs by the attack of zinc reagent

resulting in fluorinated radicals. Finally, latter species attacks at $\text{C}=\text{S}$ bond of disulfide affording the desired product **119** and the radical is regenerated. Also, in the absence of Eosin Y, but in the presence of blue light, product **119** with similar yields were observed.

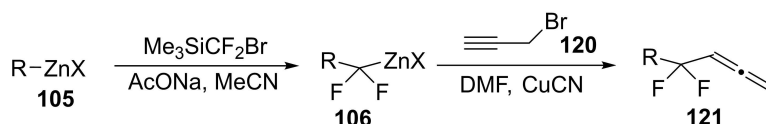
A cross-coupling reaction of difluorinated organozinc compound **106** with propargyl halides **120** in the presence of minimal quantity of copper cyanide as catalyst was disclosed by Dilman and co-workers in 2016.^[54] This allowed the easy access of 4,4-difluoroallenes **121** in excellent yields (Scheme 44).

The (phenylsulfonyl/arylthio) difluoromethylation of aldehydes **93** utilizing difluorocarbene as the $:\text{CF}_2$ source was developed by Hu group.^[55] Aromatic aldehydes having both electron-withdrawing and electron donating groups **93**, when reacted with PhSO_2Na **122** as the nucleophile and TMSCF_2Br **3** accomplished the (phenylsulfonyl)-difluoromethylation of aldehyde **123** in excellent yields (Scheme 45). It is notable that non-enolizable and enolizable aldehydes also manifested reasonable reactivity, and good to excellent yields of desired products.

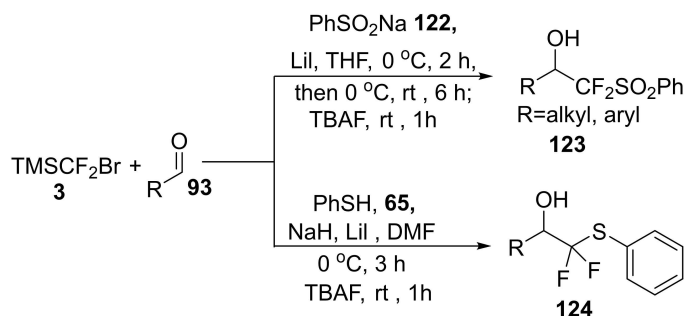
Inspired by the success in the (phenylsulfonyl)-difluoromethylation of aldehydes utilizing TMSCF_2Br **3** as the difluorocarbene source, they have studied the reactivity of (arylthio)-difluoromethylation using same protocol with ArS^- as the nucleophile (Scheme 45). The initial attempt using $n\text{-Bu}_4\text{NBr}$ in the reaction could not produce the desired outcome, prompting to utilize a Lewis acid for activating the carbonyl group, where various metal salts such as LiBr , LiCl , LiI and MgCl_2 were tested, with LiI providing the best yields. Employing NaH as base, LiI as Lewis acid, and DMF as solvent; the reaction time was optimised to 3 h at 0°C . The reaction was tested with a series of thiols **65** and aldehydes **93**, in which all of them culminated in the expected products in eminent yields. It was found that the position of substituents on the aromatic ring does not influence the yield of the reaction. Enolizable aldehydes also gave the end products in supreme yields.



Scheme 43. Copper-catalysed/light-promoted coupling of gem-difluorinated organozinc reagents with disulfides.



Scheme 44. Synthesis of 4,4-difluoroallenes.



Scheme 45. The (Phenylsulfonyl/arylthiol) difluoro-methylation.

A three-component reaction involving amines **125**, difluorocarbene **7** (generated *in situ*) and CO_2 was developed on grounds of computational predictions by Mita and co-workers in 2021.^[56] Based on a AFIR-assisted study, they envisioned the N–C and C–C bond-forming reactions of difluorocarbenes and affirmed the same by practical synthesis of difluoroglycine derivatives **126** from various tertiary amine nucleophiles, comprising pyridine, (iso)quinoline, imidazole, thiazole, and pyrazole analogs (Scheme 46). It was also found that the reaction was ineffective for ammonia.

Soon after, another three-component reaction involving fluorination followed by aminocarbonylation process was developed by Hu group (Scheme 47).^[57] This reaction provides access to α -fluoroamides **132** from readily obtainable aldehydes **93**, tertiary amine **125** and TMSCF_2Br **3** via the intermediary of difluorocarbene **7**. This multicomponent protocol facilitates an easy access to make a wide variety of

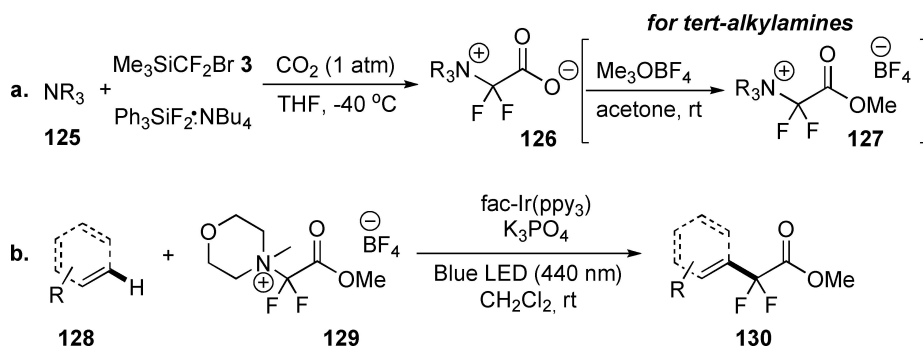


Scheme 48. Synthesis of N-difluoromethylated formidamides.

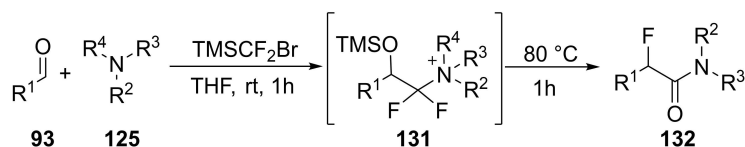
structurally different α -fluoroamides **132** by combining amide bond and C–F bond formation in a sole procedure. This reaction commences with the insertion of difluoromethylene ammonium ylides to the carbonyl carbon of the aldehydes followed by 1,2-fluorine and oxygen migrations of TMS-protected α -aminodifluoromethylcarbinol intermediates **131**.

Recently, Prakash *et al.* demonstrated a practical methodology for the preparation of N-difluoromethylated formidamides **134**.^[58] Therein, primary aryl amine **133** was treated with TMSCF_2Br **3** which acted as dual C1 Synthon (Scheme 48). The reaction was primarily carried out using BF_3 -etherate as a Lewis acid and LiOH as a base. However, since LiOH did not afford higher yields, it was replaced with Na_2CO_3 . While BF_3 -etherate resulted in good yields, likely owing to its strong attraction to fluoride, other Lewis acids like $\text{Ga}(\text{OTf})_3$ or $\text{B}(\text{OiPr})_3$ caused a significant decrease in yields. Interestingly, satisfactory product yields were achieved even in the absence of Lewis acid, which was attributed to the TMS group of TMSCF_2Br acting as a Lewis acid.

As per the proposed mechanism, difluorocarbene **7** generated from TMSCF_2Br **3**, underwent a nucleophilic attack by primary amine **133** affording an intermediate **135**, which



Scheme 46. Synthesis of α, α -difluoroglycine derivatives and photo-induced C–N bond substitution.



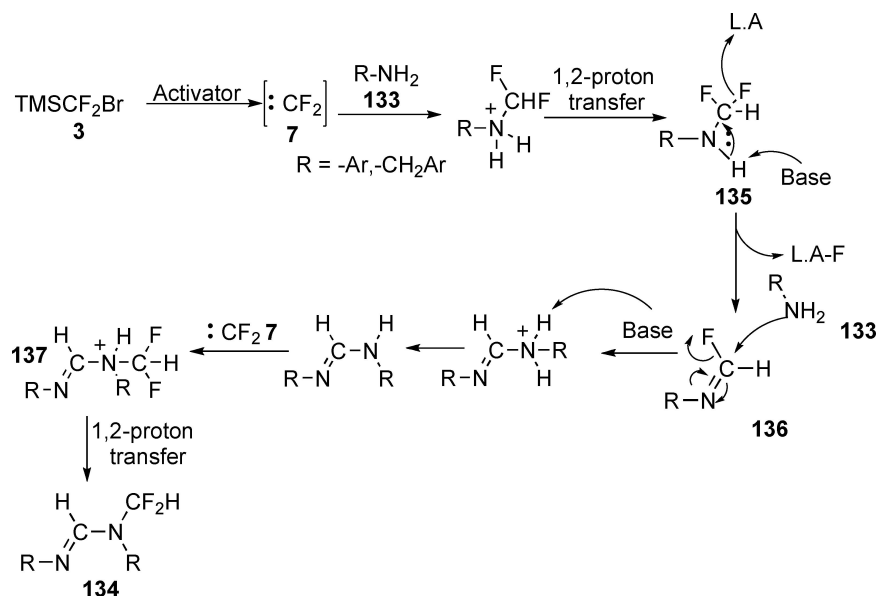
Scheme 47. Synthesis of α -fluoroamides.

in the presence of BF_3 -etherate got defluorinated to form fluoromethylimine intermediate **136**. Further addition-elimination of the latter intermediate with another amine and subsequent nucleophilic attack along with proton transfer resulted in the desired product **134** (Scheme 49). Tolerance to various functional groups and the decrease in the freon usage makes this strategy more acceptable.

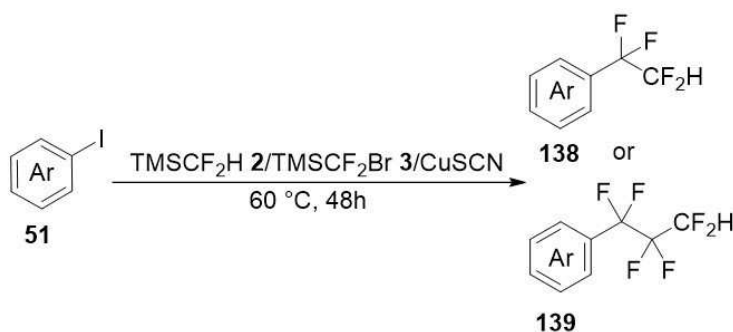
Also, it was in 2022, that Hu group came up with the path breaking highly selective and tunable, single, and double $:\text{CF}_2$ insertions into the C–Cu bond.^[59] The synthetic approach was applied to various cuprous salts, including CuI , CuCl , CuBr , CuOAc , $^i\text{PrCuOtBu}$, $^i\text{PrCuCl}$ and CuSCN . CuI exhibited the most favorable outcomes amongst the copper halides in single CF_2 insertions. However, no CF_2 insertion occurred with CuOAc , $^i\text{PrCuO}^t\text{Bu}$, and $^i\text{PrCuCl}$, presumably due to steric hindrance. Despite this, the strategy worked well with

CuSCN , resulting in notable yields and eminent selectivity. The $\text{Cu}(\text{CF}_2)_n\text{CF}_2\text{H}$ ($n=1$ and 2) formed was reacted with aryl iodides **51**, halogenation agents, and cinnamyl chloride, which opens the way for the direct attachment of HCF_2CF_2 and $\text{HCF}_2\text{CF}_2\text{CF}_2$ moieties into organic molecules such as iodobenzene. Mechanistically, the altered reactivities of $\text{Cu}(\text{CF}_2)_n\text{CF}_2\text{H}$ species having $n=0, 1, 2$ and 3 and the TMSCF_2Br **3** load governs the control of fluorocarbon chain length, that is from C_1 to C_2 and from C_1 to C_3 (Scheme 50).

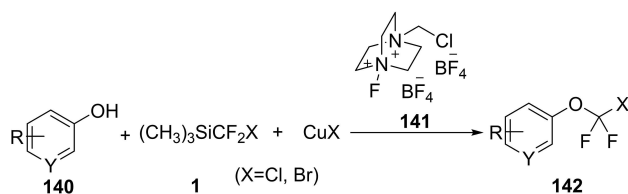
A novel method for accessing biologically relevant chloro- and bromodifluoromethyl aryl ethers **142** from phenols **140** was recently developed by Qing and co-workers.^[60] This represents the first multi-component oxidative coupling reaction involving phenols, difluorocarbene precursors $(\text{CH}_3)_3\text{SiCF}_2\text{X}$ ($\text{X}=\text{Cl}/\text{Br}$) **1** and CuCl/CuBr to synthesize the aforementioned compounds (Scheme 51). The reactions, con-



Scheme 49. Mechanistic route for the synthesis of N-difluoromethylated formamides.



Scheme 50. Single and double CF_2 insertions.



Scheme 51. Difluorocarbene-involved oxidative coupling reaction.

ducted in the presence of Selectfluor ($C_7H_{14}ClFN_2 \cdot 2BF_4$) **141**, offers mild conditions, broad substrate scope and eminent chemoselectivity.

According to the predicted mechanism, the primary desilylation of $TMSCF_2X$ with suitable nucleophile generates the halodifluoromethyl anions, which undergo α -elimination to produce difluorocarbene **7** and halide anions. The difluorocarbene **7** subsequently reacts with phenolate **143**, further oxidative coupling and reductive elimination results in the desired products **142** (Scheme 52).

Recently in 2023, Hu and co-workers were successful in developing a highly efficient multicomponent methodology involving carbonyl compounds *viz.* aldehydes or ketones **60**, amines **125** and the difluoromethylene precursor $TMSCF_2Br$ **3** for the synthesis of biologically potent α -hydroxyamides **145** (Scheme 53).^[61] The strategy was achieved by changing the reaction route of the fluoroiminium intermediate **148** from intramolecular cyclization to inter molecular hydrolysis which is facilitated by the TMS-protecting group derived from $TMSCF_2Br$ **3**. This methodology has shown an exceptional functional group compatibility and the products are obtained in good yields.

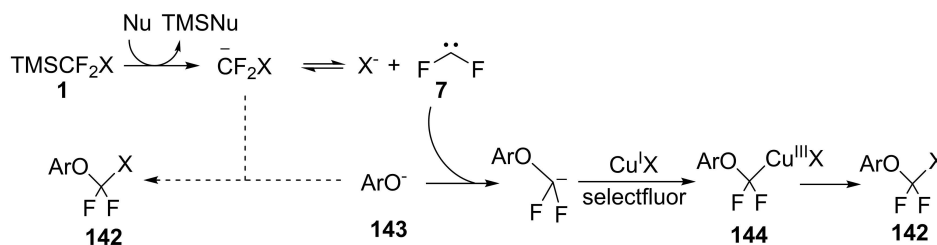
As per the proposed mechanism, the reaction began by the generation of an α,α -difluoroammonium intermediate **146**,

which is generated *in situ* from a multicomponent reaction between aldehydes, amines **125** and $TMSCF_2Br$ **3**. It then undergoes debenzoylation by bromide ion, giving intermediate **147**. The intermediate **148** is formed by the β -fluoride elimination of intermediate **147**. The latter undergoes nucleophilic addition with water. Finally, the anticipated α -hydroxyamide **145** is yielded by consequent defluorination and desilylation (Scheme 54).

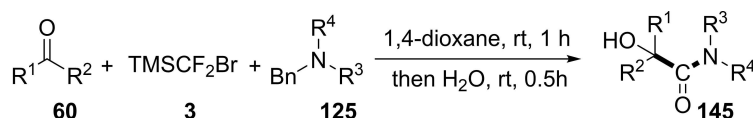
Very recently, an interesting tunable single and double difluoromethylene insertion reaction into aldehyde **93** by the intermediacy of difluorocarbene **7** from $TMSCF_2Br$ **3** was achieved by Hu group.^[62] The reaction exposed the highly selective synthesis of difluoroenolsilyl ethers **151** and tetrafluorocyclopropanolsilyl ethers **152** (Scheme 55).

The mechanism of this reaction advanced analogous to the synthesis mentioned above of α -fluoroamides (Scheme 47) reported by the same group.^[57] It involves TMS-protected quaternary ammonium salt intermediate which was engendered through a chain reaction initiated by difluorocarbene **7**. Subsequent Hofman elimination led to the products difluoroenolsilyl ethers **151** or tetrafluorocyclopropanolsilyl ethers **152** (Scheme 56).

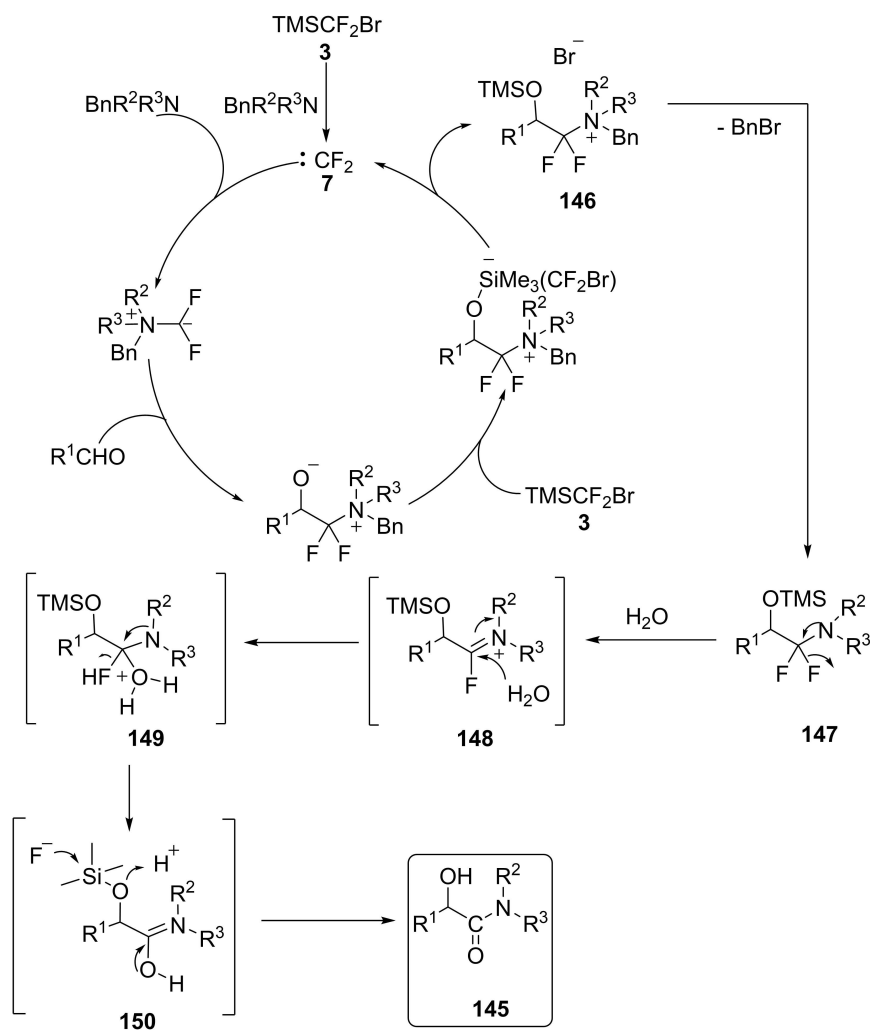
Very recently, Ye group reported an effective and simple synthesis of gem-difluoromethylenated 2-pyridones **161** involving difluoromethylene **7** through an ester insertion/acyl rearrangement reaction of 2-acetoxypyridines **159** (Scheme 57).^[63] The mechanism involves the generation of difluorocarbene **7** from $TMSCF_2Br$ **3** by the base effect. Subsequent reaction of the latter with 2-acetoxypyridines **159** results in the formation of N-difluoromethylenepyridiniumylide **160**. A final intramolecular ester insertion and acyl rearrangement afforded the gem-difluoromethylenated 2-pyridones **161**.



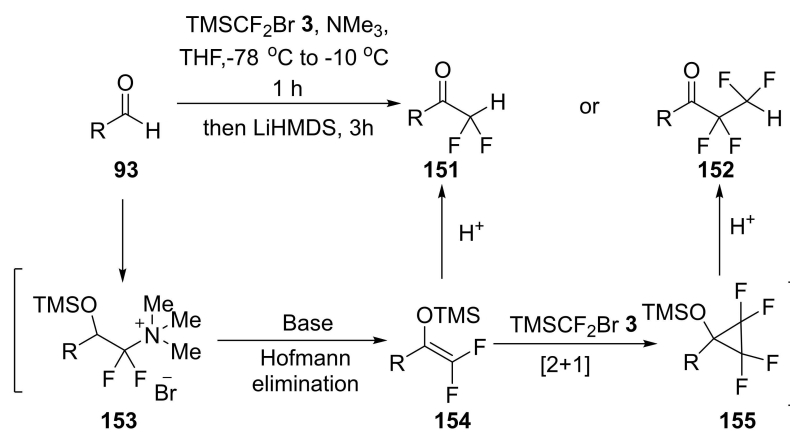
Scheme 52. Mechanism for the synthesis of halodifluoromethyl aryl ethers.



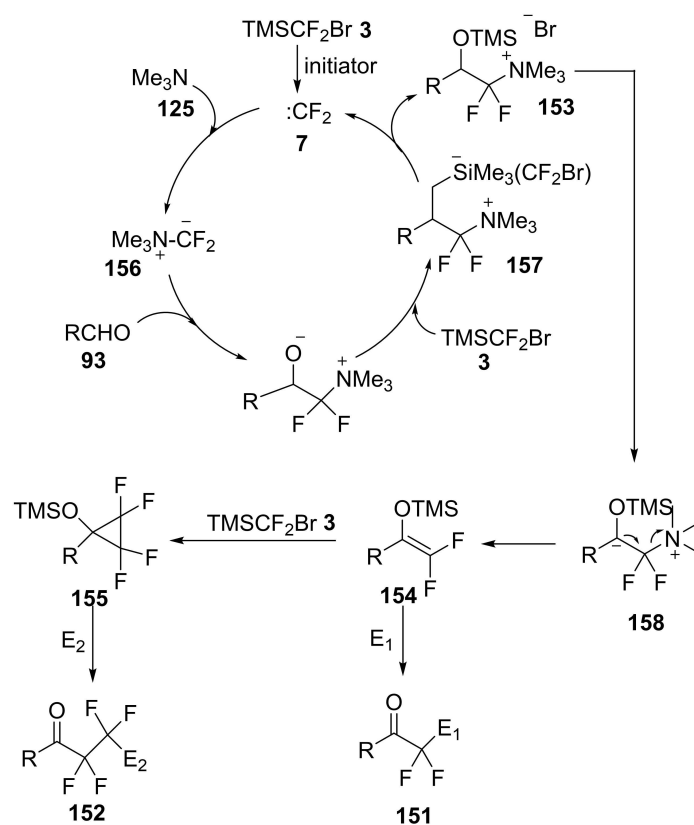
Scheme 53. Synthesis of α -hydroxyamides.



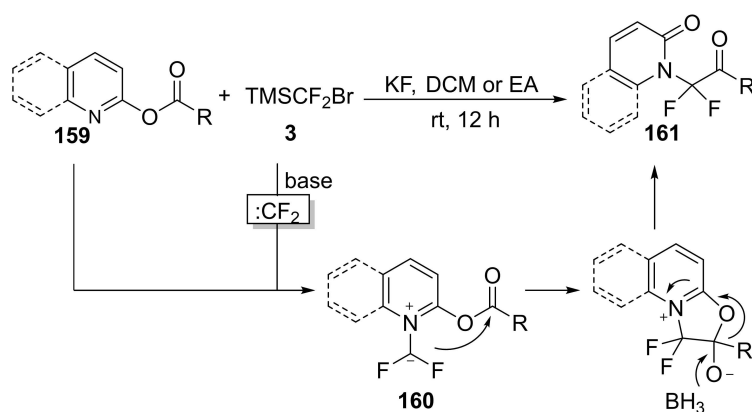
Scheme 54. Mechanism of the synthesis of α -hydroxyamides.



Scheme 55. Tunable single and double difluoromethylene insertion reaction.



Scheme 56. Mechanism of tunable single and double difluoromethylene insertion.



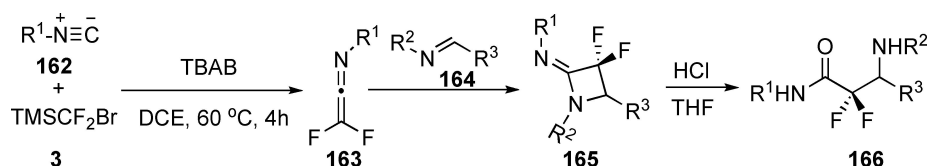
Scheme 57. Synthesis of gem-difluoromethylated 2-pyridones.

5.2. Cycloaddition Reactions

The synthetic route to α,α -difluoro- β -amino amides **166** via multicomponent reaction of TMSCF₂Br **3**, isocyanide **162** and imine **164** was reported recently (Scheme 58).^[64] The reaction proceeds through the intermediacy of difluoroketimine **163** formed by the reaction of difluorocarbene **7**

generated from TMSCF₂Br **3** and isocyanide. Subsequent [2 + 2] cycloaddition reaction of difluoroketimine **163** and imine **164** culminated in the product α,α -difluoro- β -amino Amide **166**.

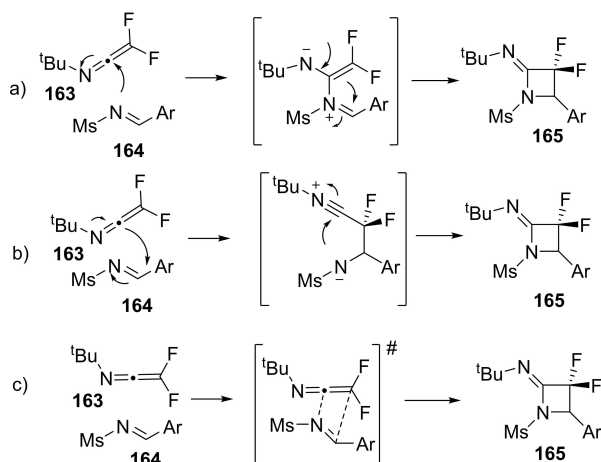
It is noteworthy that imine substrates with electron-withdrawing group yield β -lactams **165** on hydrolysis of the [2 + 2] cycloaddition products of difluoroketimines **163** and



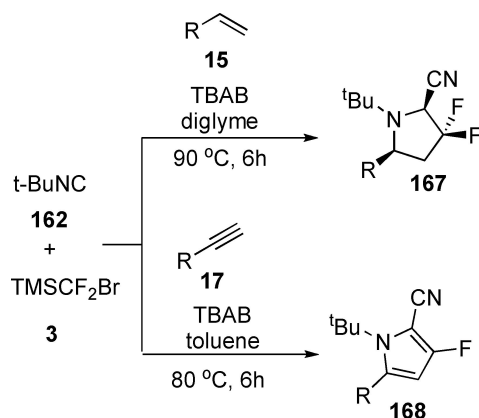
Scheme 58. Synthetic route to α -difluoro- β -amino Amides.

imines **164**. Based on the control experiments carried out and the understanding of the mechanism of Staudinger reaction, three pathways are proposed for the [2 + 2] cycloaddition process as presented in the Scheme 59.

Another recent work involves [3 + 2] cycloaddition reaction of difluoroketimines **163** generated *in situ* from isocyanide **162** and difluorocarbene **7** and alkenes **15** or alkynes **17**. The reaction afforded fluorinated pyrrolidines **167** or pyrroles **168** (Scheme 60).^[65]



Scheme 59. Plausible pathways for [2 + 2] cycloaddition.



Scheme 60. [3 + 2] cycloaddition reaction of difluoroketimines.

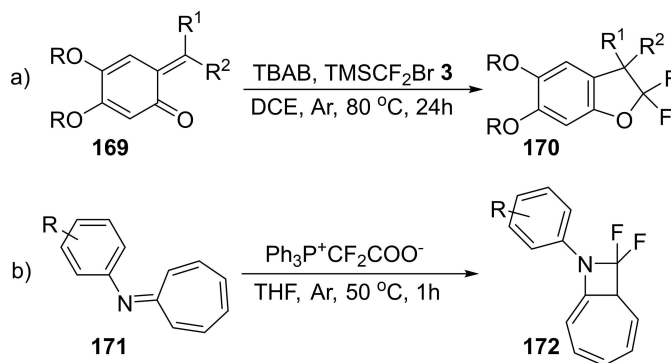
In 2021, Yang, Jiang, *et al.*, successfully disclosed a novel synthetic approach towards difluorinated heterocyclics *via* a metal-free cycloaddition.^[66] Gem-difluorinated azetidines **172** and 2,3-dihydrobenzofurans **170** were synthesized from heteroconjugated alkenes and difluorocarbenes through [8 + 1] and [4 + 1] cycloadditions, with $\text{Ph}_3\text{P}^+\text{CF}_2\text{COO}^-$ or $\text{TMSCF}_2\text{Br} **3** acting as the $:\text{CF}_2$ sources (Scheme 61). It is noteworthy that the reaction holds broad substrate scope and the products **170** and **172** are obtained with high efficiency, thereby showcasing excellent synthetic applications.$

In 2023, Yang *et al.*, successfully developed a metal-free 1,3-dipolar cyclization of C,N-cyclic azomethine imines **173** and **175** and difluorocarbene **7** (*in situ*), providing a facile access to 1,2-diazetidines **174** and **176** with excellent anticancer activity.^[67] This conversion enables the synthesis of *gem*-difluorinated four-membered heterocycles in good yield along with wide substrate scope and mild reaction conditions (Scheme 62).

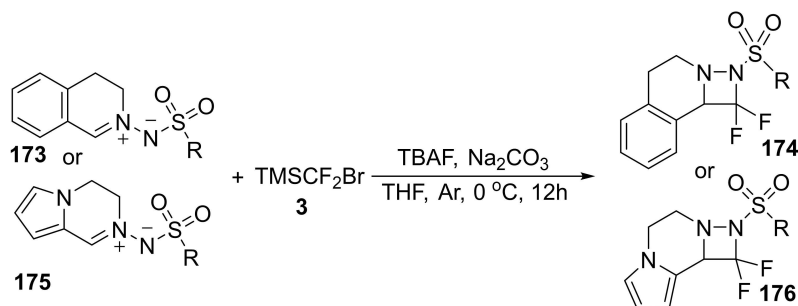
The possible mechanism involves the generation of difluorocarbene **7** by the reaction of $\text{TMSCF}_2\text{Br} **3** with TBAF followed by the addition of difluorocarbene **7** to the imine **173** to give desired product **174** (Scheme 63).$

5.3. Difluoromethylation of Alcohol and Acid

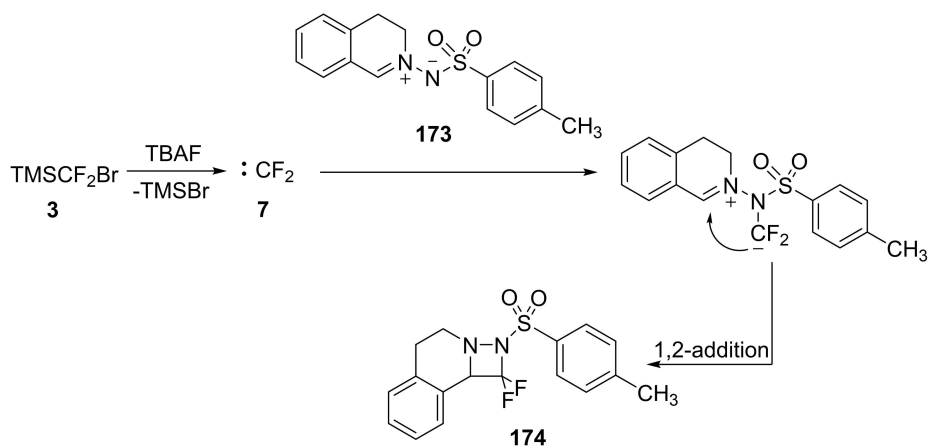
A unique difluoromethylation strategy of alcohols utilizing easily available $\text{TMSCF}_2\text{Br} **3** was developed by Hu *et al.*,$



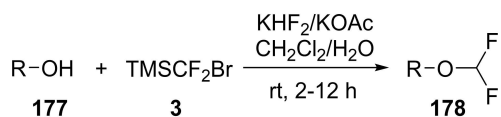
Scheme 61. [4 + 1] cycloaddition of *o*-quinone methides and TMSCF_2Br and [8 + 1] cycloaddition reaction of 8-azaheptafulvenes and $\text{Ph}_3\text{P}^+\text{CF}_2\text{COO}^-$.



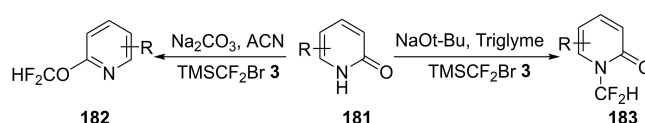
Scheme 62. 1,3-dipolar cyclization of C, N-cyclic azomethine imines and difluorocarbene.



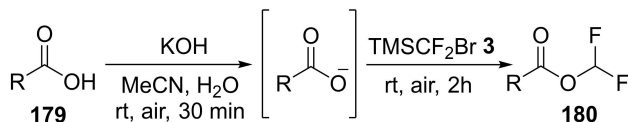
Scheme 63. Mechanism for the synthesis of 1,2-diazetidines.



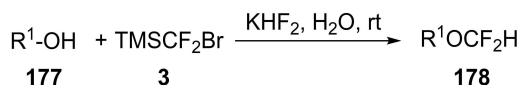
Scheme 64. Difluoromethylation of alcohols.



Scheme 67. Difluoromethylation of 2-pyridones.



Scheme 65. Synthesis of difluoromethyl esters.

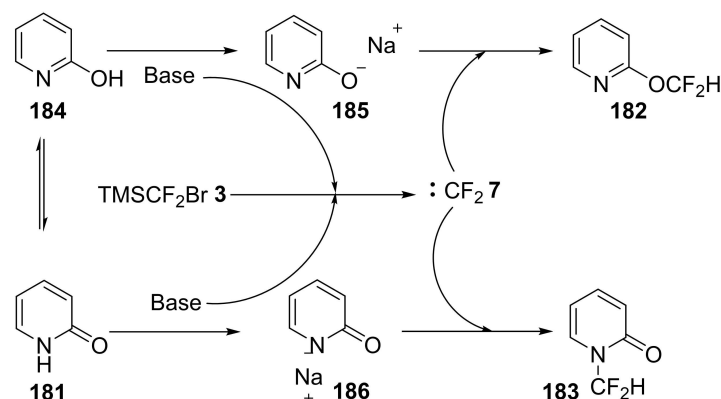


Scheme 66. Difluoromethylation of liquid alcohol.

(Scheme 64).^[68] This reaction works under weakly acidic or basic conditions, which facilitates primary, secondary and tertiary alkyl difluoromethyl ether **178** synthesis. The reaction

proceeds *via* difluorocarbene intermediate **7**. The versatility of the reaction was tested with weak bases such as KOAc and KHF₂ and also the reaction works well with almost all the alcohols. NaOH, the strong base, is found effective for alcohols with excess electrons as well as for those with less steric hindrance.

O-difluoromethylation of carboxylic acids under benchtop conditions was achieved by Prakash and co-workers in 2019 using TMSCF₂Br **3**.^[69] This aqueous base mediated approach could convert both aliphatic and aromatic acids **179** to difluoromethyl esters **180** under very mild reaction conditions. The method also finds applications in pharmaceutical sector since most drugs and biologically significant molecules contain carboxyl groups (Scheme 65).



Scheme 68. Mechanistic hypothesis for the difluoromethylation of 2-pyridones.

Liquid alcohols, whether it is primary, secondary, or tertiary effortlessly underwent the reaction with TMSCF_2Br **3** to give difluoromethyl ethers **178** in good yields at room temperature with water as solvent (Scheme 66).^[70] While in the solid alcohols **177**, the difluoromethylation proceeded with low efficiency at moderate temperature. As a complimentary and an alternate strategy, the reaction of solid alcohols **177** was carried out in water at moderate temperature by adding a minimum amount of dichloromethane to increase the solubility of alcohols, which afforded difluoromethyl ether **178** in excellent yield.

Recently, Prakash *et al.* proposed a simple strategy for the direct addition of difluoromethyl group to O- and N- of 2-pyridones, isoquinolinones and quinolines by employing TMSCF_2Br **3**.^[71] The chemoselectivity was regulated by altering parameters like solvent, temperature and base (Scheme 67).

The predicted mechanism follows deprotonation of 2-hydroxypyridine **184** and 2-pyridone **181** to produce aryloxy form **185** and amide form **186** respectively. Subsequent attack of difluorocarbene affords corresponding products **182** and **183**. Reaction was found to be tolerant towards various functional groups (Scheme 68).

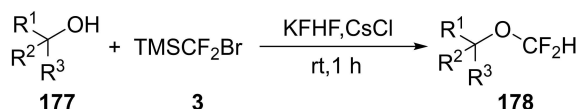
Interestingly, Bolm and co-workers proposed a synthetic approach of Difluoromethyl ethers **178** by the mechanochemical interaction of alcohols **177** and difluorocarbene **7**.^[72] The reaction was performed in a mixer mill in solvent-free setup (Scheme 69). Tertiary, secondary, and primary alcohols were found to give desired products **178** in good yields. The use of

KFHF as an activator and CsCl as a grinding auxiliary played a crucial role in the reaction. While alternative activators like KF, CsF, NaI, KOAc and TBAF as well as alkali chlorides such as LiCl, NaCl, KCl were tested, the KFHF-CsCl combination yielded the best outcome. Additionally, the reaction failed to result in the desired product when conducted with CsCl without the KFHF activator.

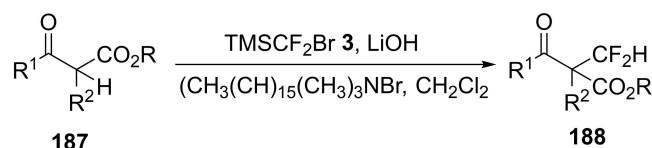
5.4. Difluoromethylation of Carbon Acids

Shibata *et al.* demonstrated a C-selective difluoromethylation of β -ketoesters **187**.^[73] The reaction was done with TMSCF_2Br **3**, LiOH and N,N,N-trimethylhexadecan-1-ammonium bromide. In this case, lithium hydroxide acts as a difluorocarbene initiator (Scheme 70). The utility of various other bases as additives were examined but LiOH proved to be the most effective. Organic bases such as DBU and DIPEA were found to be ineffective, whereas sodium and potassium based activators including NaOH, NaOMe, NaHMDS, KOH, and KOMe provided moderate yields with poor to moderate C/O selectivities. Other lithium-based reagents than LiOH, like LiCO_3 , LiOAc, n-BuLi and LiHMDS also failed to deliver favourable results. The reaction was successful with an extensive range of β -ketoesters.

Hu group reported an effective and C-selective difluoromethylation of carbon acids in the presence of TMSCF_2Br **3** and a base.^[74] Various carbon acids such as esters **191**, amides **193**, terminal alkynes, β -ketoesters, malonates,



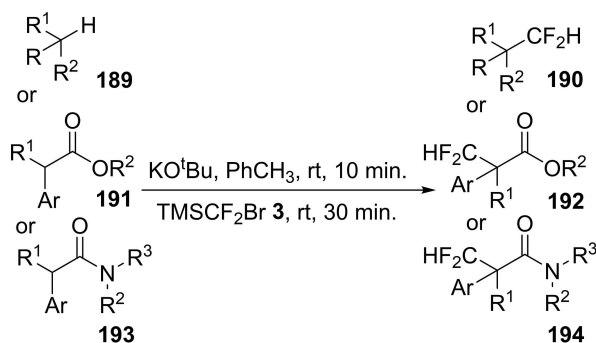
Scheme 69. Synthesis of difluoromethyl ethers.



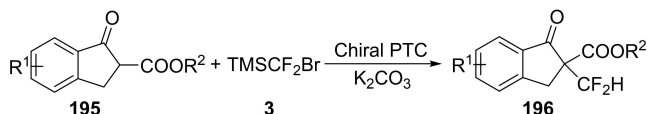
Scheme 70. C-selective difluoromethylation of β -ketoesters.

fluorenes, and other activated C–H nucleophiles, are found to give C-selective difluoromethylation products **190**, **192**, and **194** in good to excellent yields (Scheme 71). The base plays a dual role in proton abstraction and difluorocarbene generation. Among various bases evaluated, such as LiO^tBu, NaO^tBu, KO^tBu, LiHMDS, and KHMDS, KO^tBu was the sole reagent to result in the desired product. This difluoromethylation strategy is also found to be useful in modifying biologically relevant molecules such as phenylbutazone which are known to be core of pharmaceutically important molecules having anti-inflammatory, antipyretic, and analgesic activities.

An asymmetric version of the α -electrophilic difluoromethylation of β -keto esters **195** by using chiral PTC catalyst was further developed.^[75] The reaction culminated in the desired products **196** with good enantioselectivity and in high yields (Scheme 72).



Scheme 71. C-selective difluoromethylation of carbon acids.



Scheme 72. Asymmetric difluoromethylation of β -keto esters.

5.5. Ring expansion Reaction

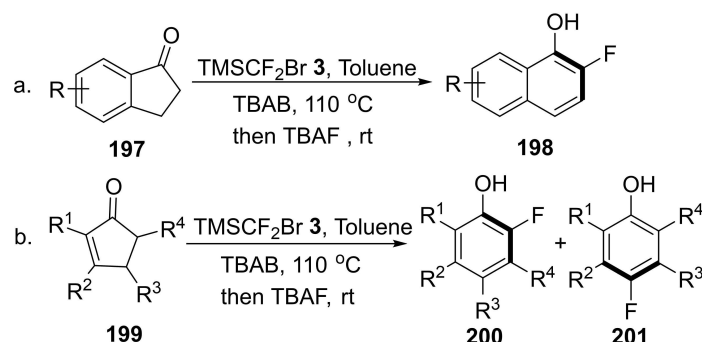
Wang and co-workers reported a straight forward synthesis of o-fluoronaphthols **198** and o/p-fluorinated phenols **200** and **201** via the reaction of difluorocarbene generated from TMSCF₂Br **3** and TBAB on indanones **197** and 2-cyclopentenones **199**.^[76] The reaction is initiated by the difluorocarbene **7** generated from TMSCF₂Br **3** (Scheme 73). Also, the methodology was extended to dihydrojasnone under the optimized reaction conditions and the isomeric products, o-fluorophenol and p-fluorophenol were obtained.

In the mechanism of the reaction, two pathways are proposed for the formation of silylenol ether intermediates **199** as shown in Scheme 74.

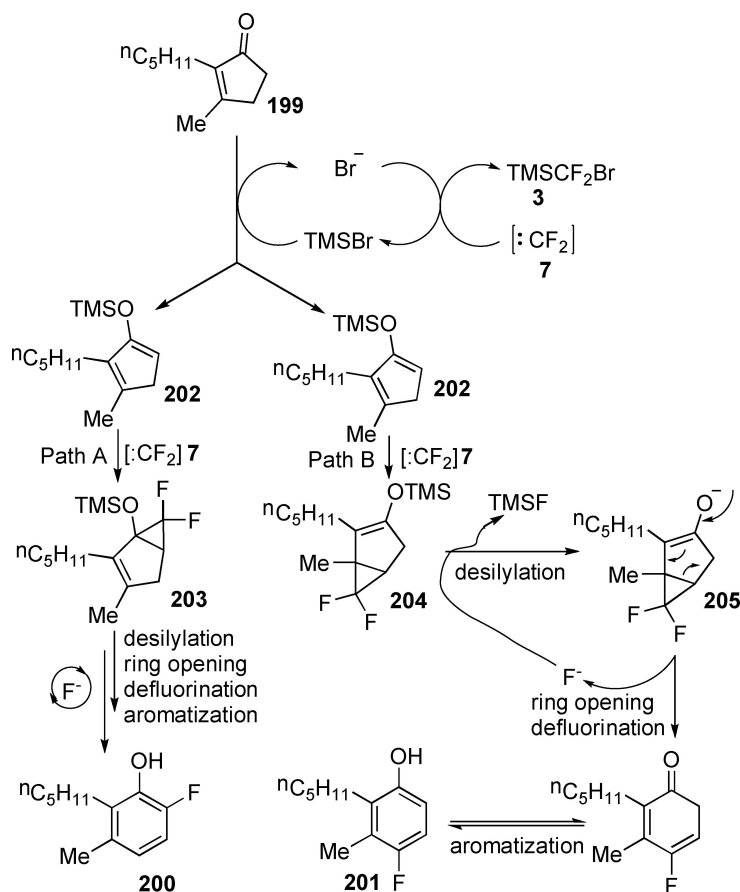
5.6. Ring Opening Reactions

In 2020, Seo and co-workers developed a simple and highly efficient approach for the deconstructive functionalization of a diverse range of unstrained cyclic systems, including, but not limited to 5- and 6- membered rings.^[77] On employing difluorocarbene **7** to generate the ammonium salt intermediates, a highly selective N-difluoromethylation takes place, followed by deconstructive halogenation and *in situ* hydrolysis, resulting in the formation of acyclic N-formyl haloamines **207** (Scheme 75). This protocol highlighted the straightforward modification of azacyclic cores in complex molecules containing versatile functional groups. Also, this offers a well-defined pathway for diverse scaffolds that cannot be formulated via usual von Braun-type reactions. Apart from its efficacy, functional group tolerance, chemo- and regioselectivity, this last stage diversification process to generate diverse scaffolds with potent biological activity is a defining feature.

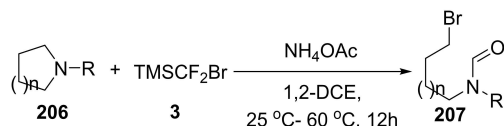
The ring opening difluoromethylation halogenation of cyclic ethers and thioethers **208** was achieved by using TMSCF₂Br **3** or TMSCF₂Cl **5**, which is the difluorocarbene source generated in presence of KHF₂ and LiCl/LiBr (Scheme 76).^[78] This method is efficient with a wide variety of



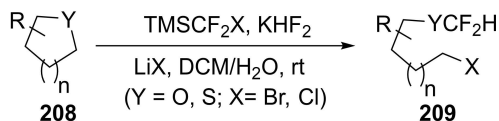
Scheme 73. Synthesis of naphthols and o/p-fluorinated phenols.



Scheme 74. Mechanism of naphthols and *o/p*-fluorinated phenol syntheses.



Scheme 75. Ring-opening halogenation via N-difluoromethylative cleavage of C–N bond.



Scheme 76. Ring opening difluoromethylation halogenation.

cyclic ethers and thioethers **208** and it also showcased the modification of natural products such as *p*-coumaric acid, formononetin, Estrone and Tocopherol.

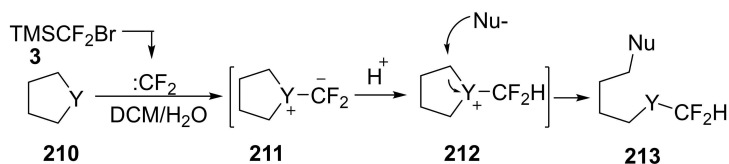
Mechanistically, the more Lewis basic oxygen or sulphur reacts to generate difluoromethyleneoxonium/sulphonium ylide **211**. The latter undergoes protonation to give a difluorometh-

ylidialkylxonium intermediate **212**, which would undergo a nucleophilic attack at one of the alkyl substituents, yielding difluoromethyl ether **213** (Scheme 77).

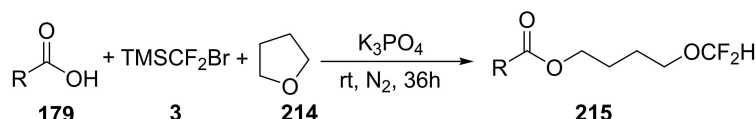
In 2022, Song and co-workers reported a novel strategy for the synthesis of difluoromethyl ethers **215** by difluorocarbene-mediated C–O bond dissociation of cyclic ethers **214**.^[79] This facile entry was applicable for a wide range of substrates such as hydroxyl, alkynes, and olefins (Scheme 78). This method is of greater importance in pharmaceutical industries due to its high chemoselectivity and late-stage modification.

During the same period, Qiu and co-workers disclosed a Cu catalysed two-fold C–O bond separation of cyclic ethers **214** with carboxylic acids **179**, NaI, and TMSCF₃ **1** to access broad range of structurally divergent iodoalkyl esters **216** with exceptional functional group tolerance.^[80] This approach towards synthesis of iodoalkyl esters **216** enhanced synthetic and process efficiency (Scheme 79).

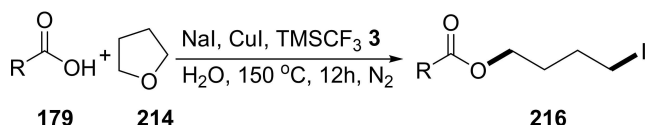
Xie and Sheng recently developed a straightforward protocol for the difluoromethoxybutylation of *N*-aryl-*N*-hydroxylamines **218**.^[81] They utilized TMSCF₂Br **3** as the source of difluorocarbene and tetrahydrofuran serve as both a solvent



Scheme 77. Mechanism of ring opening difluoromethylation halogenation.



Scheme 78. Synthesis of difluoromethyl ethers.



Scheme 79. Synthesis of highly functionalized iodoalkyl esters.

and a reactant, and the reaction was conducted in the presence of air without the use of transition-metal catalysts (Scheme 80). The process involves a novel ring-opening reaction of THF induced by difluorocarbene **7**, enabling the difluoromethoxybutylation of *N*-aryl-*N*-hydroxylamines **217**. Mechanistic studies demonstrated that difluorocarbene **7** activates THF intermolecularly, forming an oxonium ylide which subsequently undergoes nucleophilic attack by *N*-aryl-*N*-hydroxylamines **218**.

5.7. Cyclopropanation Reactions

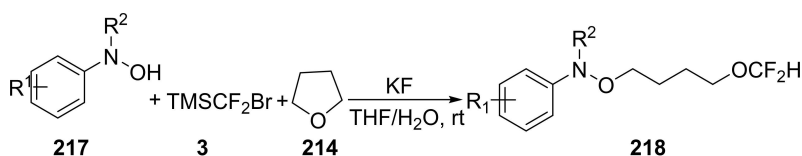
In 2013, Hu and co-workers introduced a highly effective protocol for the difluoromethylation of C–C multiple bonds and difluoromethylation of heteroatom nucleophiles with TMSCF_2Br **3** as the single $:\text{CF}_2$ **7** source (Scheme 81).^[19c] This simple method enables the facile preparation of gem-difluorocyclopropanes **16**/difluorocyclopropenes **18** and difluoromethylated compounds containing O-, S-, N-, and P **220**. At a higher temperature (110 °C), bromide catalyzes the fragmentation of TMSCF_2Br **3** leading to difluorocarbene

intermediate. The latter is reactive towards alkenes **15**, and both the electron-rich and electron-poor alkynes **17**. Fragmentation of hydroxyl ion at low temperature (0 °C) led to the difluoromethylation of thiophenols with outstanding tolerance for functional group. The difluoromethylation of thioalcohols, sulfonates, heterocyclic amines and even hydrophosphine oxides was also achieved.

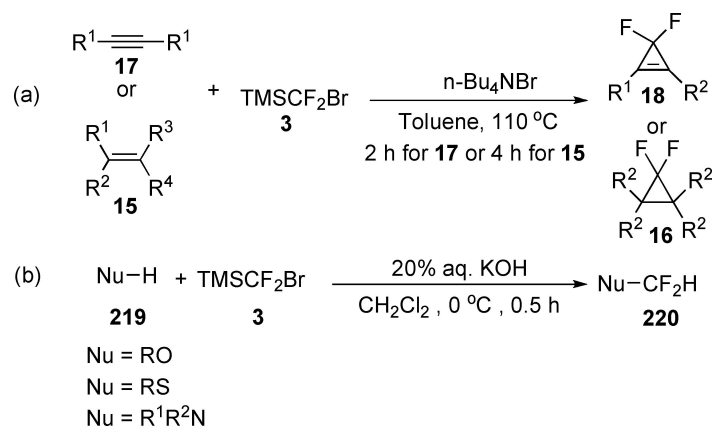
Wang, Zhu, and co-workers devised a metal facilitated well-regulated ring opening of siloxydifluorocyclopropanes **222** to α -gem-difluorocyclopentenones **223**. Using $n\text{Bu}_4\text{N}_4\text{Br}$ as the catalyst and TMSCF_2Br **3** as a unique difluorocarbene source, difluorocyclopropanation of methyl vinyl ketones containing a β -alkylthio group was achieved. The resulting Copper difluorohomoenolates was further transformed into α -gem-difluorocyclopentenones **223** via an intramolecular addition and elimination reaction (Scheme 82).^[82]

As per the proposed mechanism, the siloxydifluorocyclopropanes **222** were obtained by the TBAB-catalyzed reaction of α -acyl ketene dithioacetals and TMSCF_2Br **3**. Subsequent treatment with CuCl and β -carbon elimination results in a copper difluorohomoenolate **226**, which undergoes an intramolecular $\text{S}_\text{N}2$ type reaction to afford the desired product **223**. The above mentioned one-pot synthesis was also applied to convert vinylogous thioesters using silver difluorohomoenolates (Scheme 83).

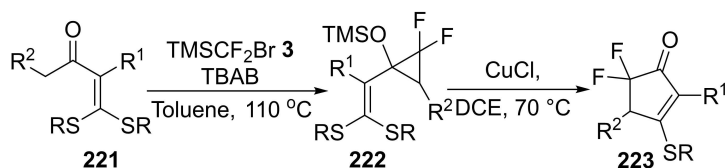
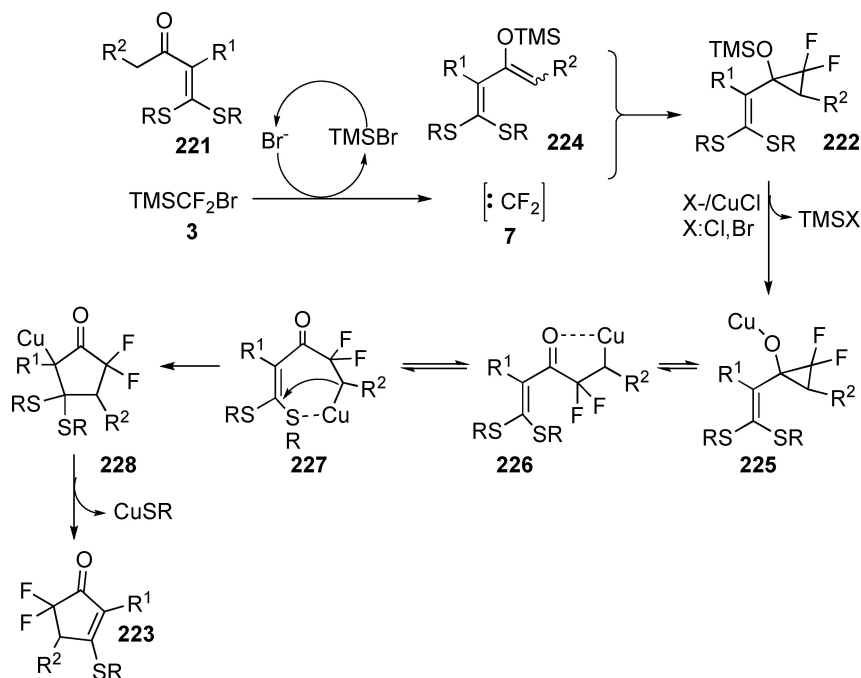
Hu and Liu described the foremost synthetic protocol for aryl perfluorocyclopropyl ethers **230**.^[83] The synthesis involved a [2 + 1] cyclopropanation of aryl trifluorovinyl ethers **229**



Scheme 80. Synthesis of difluoromethoxybutylation.

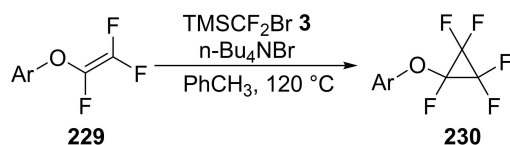


Scheme 81. Preparation of gem-difluorocyclopropanes/difluorocyclopropenes.

Scheme 82. Synthesis of α gem-difluorocyclopentenones.Scheme 83. Mechanism of α gem-difluorocyclopentenones synthesis.

(TFVE) and TMSCF₂Br **3** (Scheme 84). This cycloaddition reaction of two fluorine embedded reactants proceeded efficiently in the presence of a catalytic amount of *n*-Bu₄NBr

in toluene at reflux conditions. The reaction was observed to be compatible with both electron-rich and electron-deficient aryl TFVEs and it also performed effectively with aryl TFVEs



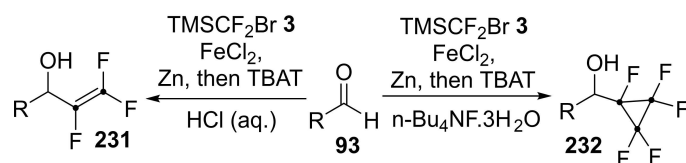
Scheme 84. Synthesis of aryl perfluorocyclopropyl ethers.

possessing a range of functional groups. Furthermore, the reaction was successfully expanded to aryl trifluorovinylthioethers in which they were rapidly converted into aryl perfluorocyclopropylsulfone *via* [2 + 1] cycloaddition and the successive oxidation reaction.

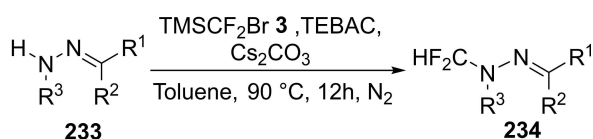
Also, Hu and colleagues recently introduced a direct method for trifluorovinylation and pentafluorocyclopropylation of aldehydes **93** using TMSCF₂Br **3** (Scheme 85).^[84] This approach relied on TMSCF₂Br-mediated controllable fluorocarbon chain elongation (CFCE), in which TMSCF₂Br **3** acted as the only source of fluorocarbon. It generated two key intermediates- a TMSCF₂ radical and a difluorocarbene. When TMSCF₂ radical was employed, trifluorovinylation of aldehydes was achieved with a CFCE from C1 to C2 whereas difluorocarbene enabled the pentafluorocyclopropylation **232** with a CFCE from C1 to C3. The process was initiated with the homo-coupling between TMSCF₂Br **3** and zinc dust, leading to the formation of TMSCF₂CF₂TMS.

5.8. N-Difluoromethylation Reactions

An interesting method for the preparation of *N*-difluoromethylhydrazones **233** *via* base-promoted *N*-difluoromethylation was reported by Wu group in 2019 (Scheme 86).^[85] This method utilized hydrazones with N–N bonds as chain substrates **233** and TMSCF₂Br **3** as difluoromethylation reagent. The reaction was conducted in the presence of a phase



Scheme 85. TMSCF₂Br-enabled trifluorovinylation and pentafluorocyclopropylation.



Scheme 86. Synthesis of *N*-difluoromethylhydrazones.

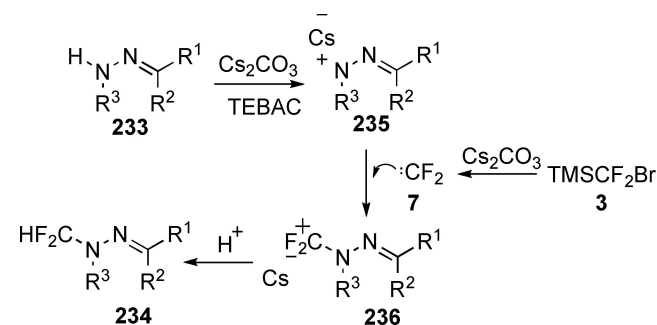
transfer agent (TEBAC), in toluene under N₂ atmosphere and afforded an array of difluoromethylation products **234** in good yields, which exhibit good stereoselectivity and excellent functional-group-tolerance.

According to the suggested mechanism, the reaction commenced with the conversion of hydrazone into intermediate **235** on treatment with Cs₂CO₃ and TEBAC. Further, intermediate **235** was attacked by :CF₂ carbene, which was produced from TMSCF₂Br **3** with Cs₂O₃ as a promotor, to give intermediate **236**. The final product was then obtained *via* the protonation of intermediate **236** (Scheme 87).

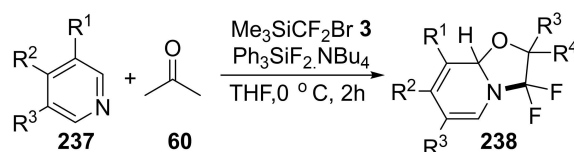
In 2022, Hayashi and co-workers predicted a synthetic strategy of α , α -difluorinated *N*-heterocyclic compounds **238** computationally using artificial force induced reaction.^[86] Later, the multicomponent reaction involving difluorocarbene was developed experimentally by the cycloaddition of an azomethine ylide with various coupling partners like imines, alkenes, aldehydes, alkynes, and ketones **60** (Scheme 88). Pyridine was identified as the major substrate for the generation of ylide intermediate in this approach. Moreover, the potent applications of this method in drug discovery makes it more significant.

5.9. Difluorohomologation Reactions

The halogenative difluorohomologation of ketones **239** initially employing silylenol ethers as the substrates was developed in 2015 by Dilman and co-workers (Scheme 89).^[10c] Later on, the whole process was performed with ketones to avoid the time delay in isolating and purifying enol ethers.



Scheme 87. Mechanism of *N*-difluoromethylhydrazones synthesis.



Scheme 88. Synthesis of α , α -difluorinated *N*-heterocyclic compounds.

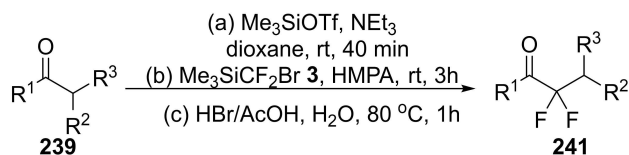


Scheme 89. Difluorohomologation reaction.

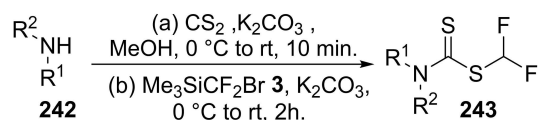
The reaction involved silyl enol ether formation followed by difluorocarbene addition using TMSCF_2Br **3**, and finally the halogenation of the intermediate cyclopropanes with either N-bromo- or N-iodosuccinimide. It was observed that the efficiency of the process was dependent on the difluorocarbene addition step. Also, the entire process was conveniently performed without the isolation of intermediates. This halogenative difluorohomologation was found to be effective for cyclic and acyclic ketones affording the desired products **240** in good to excellent yields.

The products obtained were found to open up an easy entry to fluorine substituted heterocycles. That is, the treatment of homologated products with phenylhydrazine in presence of base K_2CO_3 yielded either pyrazoline or pyrazole by varying the reaction conditions. The reaction of hydride or a Grignard reagent with the carbonyl group of the homologated product gave its alcohol derivative. The latter underwent cyclization leading to *gem*-fluorinated oxetanes under basic conditions.

Also in the same year, they have reported another difluorohomologation of ketones **239**.^[10d] The first step of this reaction is the silylation followed by difluorocyclopropanation led to cyclopropane ring (Scheme 90). The latter undergo ring opening affording difluorohomologation product of the ketone **241**. The entire process is carried out in one pot.



Scheme 90. Difluorohomologation of ketones.



Scheme 91. Synthesis of S-difluoromethyl dithiocarbamates.

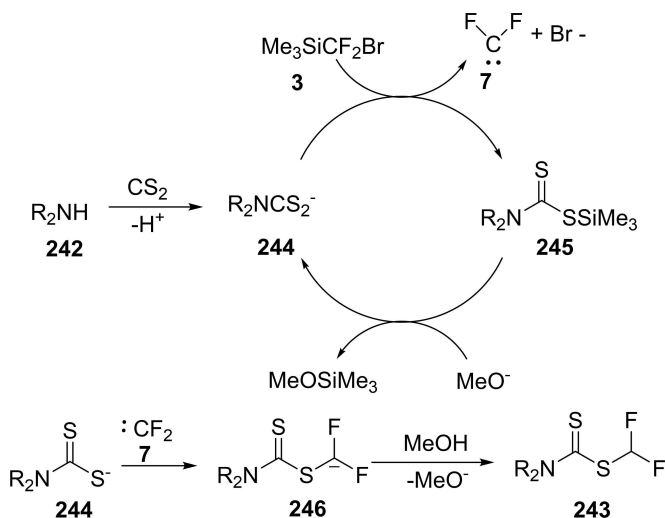
5.10. Miscellaneous Reactions

In 2015, Dilman *et al.*, developed a robust method for the synthesis of S-difluoromethyl dithiocarbamates **243** from secondary amines **242**.^[87] The protocol utilizes (bromodifluoromethyl) trimethylsilane as the difluorocarbene source (Scheme 91).

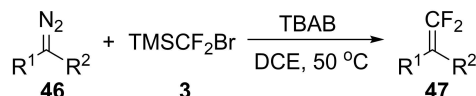
According to the proposed mechanism, the dithiocarbamate anion acts as a Lewis basic promotor towards the formation of difluorocarbene **7** and an intermediate having an S–Si bond **245**. Following the desilylation of the latter, dithiocarbamate anion **244** is regenerated and eventually reacts with the difluorocarbene **7**, leading to the formation of final product **243** (Scheme 92).

Wang group disclosed an analogous process for the synthesis of 1,1-difluoroolefins **47** by using TMSCF_2Br **3** as the difluorocarbene source and tetrabutylammonium bromide (TBAB) as the initiator from diaryl diazomethane **46** (Scheme 93).^[88] Similar to Hu's work^[31] this reaction also showed a broad substrate scope for diaryl diazomethane **46**, and diazo compounds having electron-withdrawing ester groups.

Dilman and co-workers reported the generation of difluorinated phosphorus ylides $\text{PPh}_3=\text{CF}_2$ by the combination of $\text{Me}_3\text{SiCF}_2\text{Br}$ **3**, PPh_3 **10** and DMPU under mild conditions.^[89]



Scheme 92. Proposed mechanism for S-difluoromethyl dithiocarbamates synthesis.



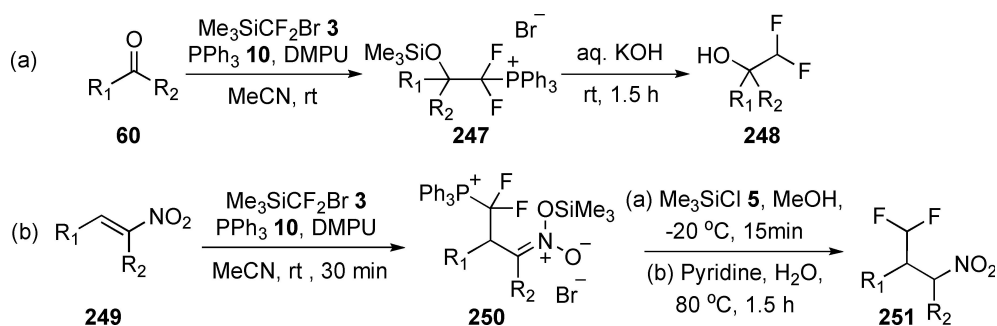
Scheme 93. Synthesis of difluoroolefins.

This methodology was utilized for the difluoromethylation of ketones **60** and nitro alkenes **249**, and in each case the corresponding difluoromethylated products **248** and **251** were delivered. Notably, the substrates **60** and **249**, traditionally considered challenging for the addition of fluorinated carbanions, exhibited improved reactivity in this case. This is likely due to the Lewis acidic activation by a silylium species generated during the formation of the phosphorus ylide, which enhanced the substrates' susceptibility to nucleophilic attack (Scheme 94).

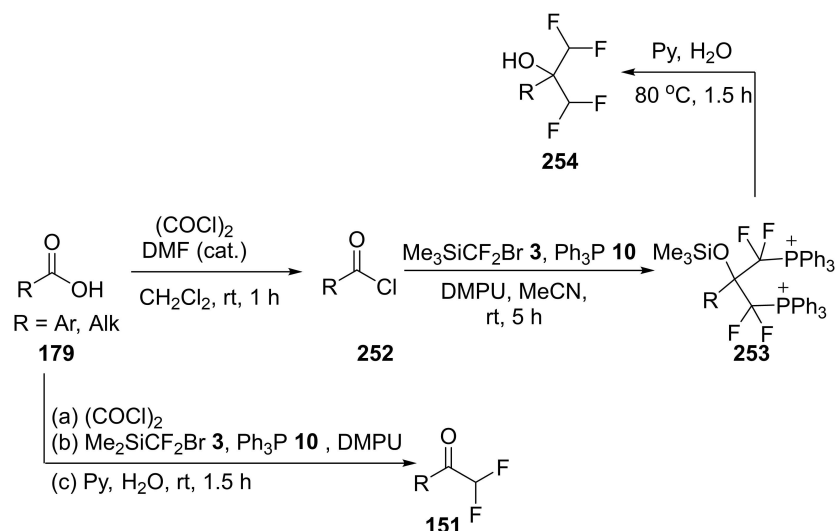
Also, the same group described a single step approach for the difluoromethylation of carboxylic acids **179**.^[90] The reaction involves the interaction between acid chloride **252**, the intermediate formed from the carboxylic acid and difluorinated phosphorus ylides generated in situ from TMSCF₂Br **3** and PPh₃ **10**. Based on the reaction conditions, bis(difluoromethyl) substituted alcohols **254** or difluorinated ketones **151** are obtained. However, the bulky carboxylic acids yield ketones as single products (Scheme 95).

A systematic approach for the selective introduction of CF₂H motif into β -keto amides **255** using TMSCF₂Br **3** was developed by Wang *et al.*, in 2021.^[91] This strategy allows facile access to a wide range of α -difluoromethyl β -keto amides **256**, **257**, and **258** with high yields and excellent Carbon/Oxygen regioselectivities under mild reaction conditions (Scheme 96). The C/O selectivity of β -keto amides can be easily modulated by merely switching the base. It was also possible to reduce the obtained products into amino alcohols containing CF₂H motifs. On top of that, the first enantioselective electrophilic difluoromethylation of β -keto amides **257** has been successfully accomplished using phase-transfer catalysis.

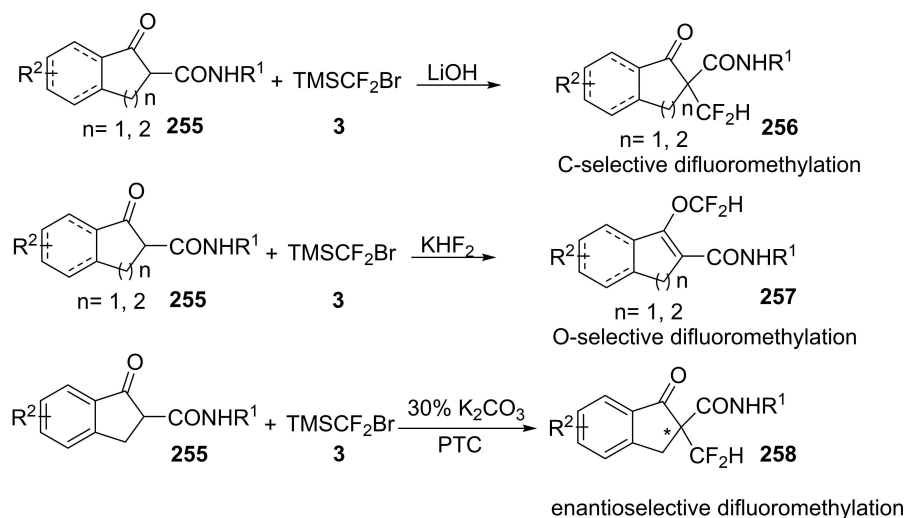
Possible mechanisms for both C-difluoromethylation and O-difluoromethylation of amides were predicted and it indicates the dual role of LiOH as a difluorocarbene activator and as a O-steric blocking site (Scheme 97). Moreover, a novel protocol for the asymmetric difluoromethylation of β -keto



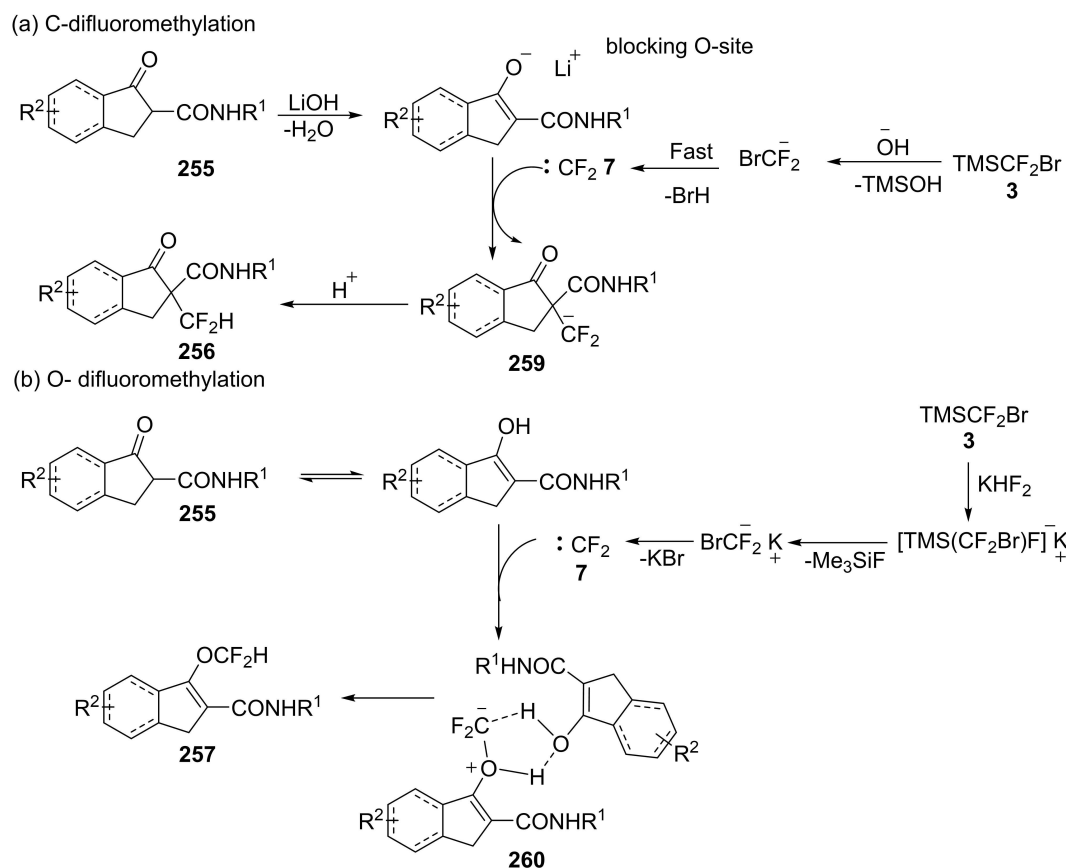
Scheme 94. Nucleophilic difluoromethylation.



Scheme 95. Difluoromethylation of carboxylic acids.



Scheme 96. Difluoromethylation of β -keto amides using TMSCF_2Br .



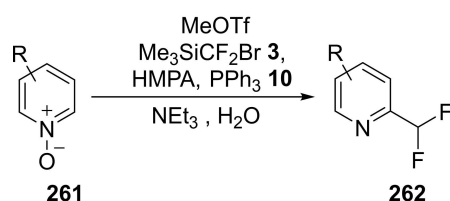
Scheme 97. Mechanism for selective introduction of CF_2H motif.

amides **255** using phase transfer catalysts derived from cinchona alkaloid was also described.

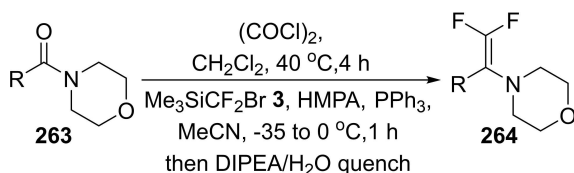
A new synthetic approach towards difluoroalkyl-substituted heteroarenes **262** was successfully demonstrated by Dilman

and Trifonov in 2021.^[92] These diverse fluorinated compounds were achieved through one-pot synthesis using a three-component reaction involving N-methoxyazinium salts **261**, difluorocarbene **7** and electrophilic or radical reagents (Scheme 98). The formation of difluorinated Phosphonium salt was recognized as the key step in the reaction. Interestingly it was found that the reaction works well with N-oxides of pyridine, quinoline and isoquinoline. Also, it is worthy to mention that the above reaction exhibits excellent C2 selectivity.

Dilman and Trifonov reported a metal-free one-pot conversion of amides **263** into gem-difluoro enamines **264**.^[93] The olefination worked well with various amides and was associated with a low temperature nucleophilic dephosphorylation of addition intermediate. Further, the obtained



Scheme 98. Difluoromethylation of Azine N-Oxides.



Scheme 99. Gem-difluoroolefination of morphine amide.

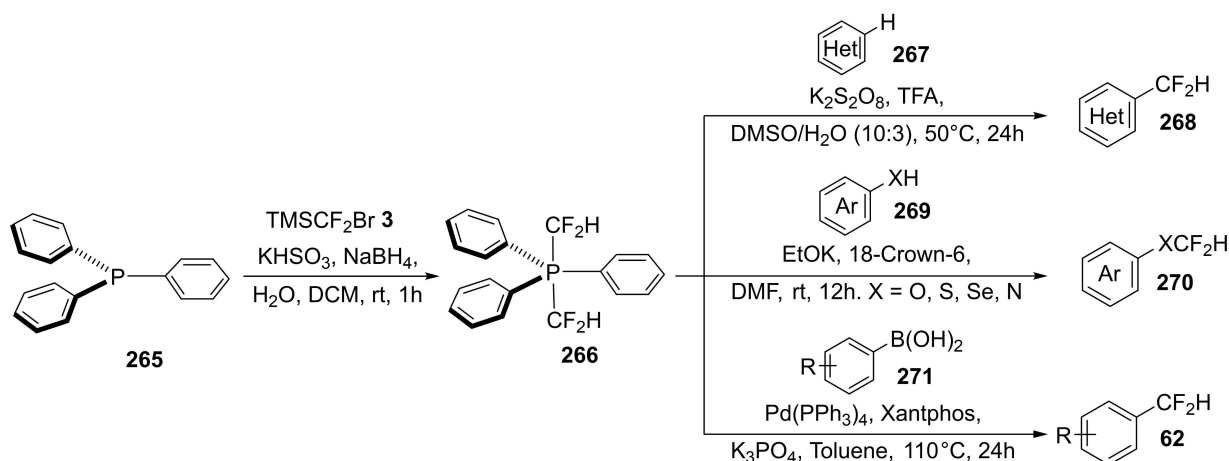
difluorinated enamines **264** were also transformed into α -fluoroalkylated amines (Scheme 99).

Very recently, Hu and co-workers reported a facile and operationally simple method for the preparation of bis(difluoromethyl) pentacoordinate phosphoranes **266** (Scheme 100).^[94] These reagents serve as precursors for the generation of difluorocarbene (:CF₂) **7** and difluoromethyl radical (.CF₂H) intermediates, enabling diverse fluoroalkylation reactions (Scheme 100). On employing difluoromethyl radicals (.CF₂H), several key transformations were achieved, which includes oxidative difluoromethylation of electron-deficient heterocycles **267**, photoredox-catalyzed difluoromethylation of alkene and nickel/photoredox dual-catalyzed difluoromethylation of aryl bromides (Scheme 104). Furthermore, the difluorocarbene (:CF₂) **7** has facilitated reactions namely gem-difluorocyclopropanation of alkenes, base-catalyzed difluoromethylation of heteroatom nucleophiles **269**, Pd-catalyzed difluoromethylation of arylboronic acids **271** and copper-mediated trifluoromethylation of aryl iodides. These methods have made possible late-stage fluoralkylation of drug molecules, making the approach a practical tool for drug discovery and manufacturing.

The future of difluorocarbene chemistry lies in improving the selectivity, efficiency and sustainability of the reactions, particularly in drug discovery. The integration of modern synthetic strategies like flow chemistry will lead to the commercialization of the products.

6. Conclusion

The enormous applications of organofluorine compounds particularly in drugs and agrochemicals triggered the interest of chemists in developing fluorination strategies. Among the plethora of methods developed reactions with difluorocarbene



Scheme 100. Synthesis of bis(difluoromethyl) pentacoordinate phosphoranes and its fluoroalkylation reactions.

generated from TMSCF_3 and its derivatives, TMSCF_2Br and TMCF_2Cl occupy a distinguished position being a facile and cost-effective method. The introduction of difluoromethyl (CHF_2^-), fluoromethanylylidene ($-\text{CF}=\text{}$), difluoromethylene ($-\text{CF}_2-$), and difluoromethylidene ($\text{CF}_2=\text{}$) moieties into targeted molecules is achieved through the intermediacy of difluorocarbenes. The examples presented in this review portray the developments in the difluorocarbene chemistry as a synthetic tool. We hope that this review will be of educational value in understanding the reactions involving difluorocarbene and it is conceivable that difluorocarbene chemistry will continue to be explored as a versatile fluorination protocol.

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Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability Statement

This is a review article. No research data are provided.

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